



Permit Audit Workflow
On-site Digital Verification



SYSTEM PORTAL INTERFACE

- Dashboard with Permit Logs
- Application Status Tracking
- Action & Update Buttons



Application
Status



Action &
Update

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ABSTRACT

Rapid urbanization in Mogadishu has increased demand for efficient, transparent municipal governance, particularly in building permit administration. The conventional permit process managed by the Banadir Regional Administration relies on manual, in-person procedures, resulting in processing delays, limited transaction visibility, and difficulty verifying payments and documents. Existing digital permit systems in literature largely target desktop-based administration and rarely integrate mobile citizen access, automated payment verification, and real-time field authentication within one platform, leaving a practical gap for municipalities such as Mogadishu that need an accessible, low-infrastructure solution. To address this gap, this study aimed to design, implement, and evaluate a synchronized Web and Mobile Building Permit System for Mogadishu Municipality. Using an Agile development methodology, the system was implemented as a multi-tier client-server architecture comprising a React.js web administration console, a cross-platform Flutter mobile application, and a Node.js/Express.js RESTful API connected to a MongoDB database, with core features including automated appraisal fee calculation, role-based access control (RBAC), JWT authentication, and camera-based QR code verification. The system was evaluated through black-box functional testing, mobile unit and integration testing, hardware compatibility testing on five Android devices, and security testing against common web vulnerabilities. Across 38 test cases spanning these four dimensions, all scenarios passed with no critical defects recorded, confirming reliable application processing, secure transaction handling, and on-site permit authenticity verification via QR code scanning. These findings indicate that the system reduces reliance on manual procedures and strengthens transparency and accountability in municipal permit administration. The study

contributes a scalable technical model for digitalizing local government services in resource-constrained environments such as Somalia, with implications for future integration of GIS-based verification and offline-capable field tools.

Keywords: Building Permit System; Digital Governance; Municipal Public Administration; QR Code Verification; Role-Based Access Control (RBAC); RESTful API.

DEDICATION

I dedicate this research work to Almighty Allah for granting me the divine guidance, knowledge, mental clarity, and resilience throughout my academic journey and for enabling the successful completion of this study.

I express my deepest dedication to my beloved parents, whose unconditional love, sacrifices, continuous prayers, and unwavering faith in my potential have been the cornerstone of my academic accomplishments. Their endless support and encouragement inspired me to persevere through every technical challenge and pursue excellence.

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Finally, I dedicate this work to all future researchers, software engineers, and urban technology innovators dedicated to modernizing public sector administration and civic governance through digital transformation. It is my hope that this study will serve as a valuable reference for building secure, transparent, and efficient municipal platforms in Somalia and beyond.

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LIST OF ABBREVIATION

Abbreviation	Meaning
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
CSS	Cascading Style Sheets
DB	Database
GIS	Geographic Information System
HTTP / HTTPS	Hypertext Transfer Protocol / Secure
IP	Internet Protocol
JSON	JavaScript Object Notation
JWT	JSON Web Token
MERN	MongoDB, Express.js, React, Node.js
MVC	Model-View-Controller
SDLC	Software development life cycle
PDF	Portable Document Format
QR Code	Quick Response Code
RBAC	Role-Based Access Control
REST	Representational State Transfer
UI	User Interface
URL	Uniform Resource Locator
UX	User Experience

CHAPTER I: INTRODUCTION

1.1 Background of the Study

The digital transformation of municipal administrative services has emerged as an essential foundation for modern urban planning, efficient governance, and sustainable spatial development. Globally, traditional building permit systems are transitioning from isolated paper workflows toward integrated digital governance to eliminate processing delays, administrative friction, and operational opacity (Fauth et al., 2026). Across developing regions, municipal administrations are adopting centralized web and mobile platforms to enhance public service accessibility and streamline document verification (Thapaliya & Esichaikul, 2025).

In Somalia, rapid urbanization and post-conflict reconstruction have positioned Mogadishu as a fast-growing metropolitan center, generating a substantial volume of building permit applications across the Banadir Regional Administration (BRA) (Ahmed et al., 2025).

Despite this construction surge, local permitting relies on isolated counter-entry software and manual routines. Property developers must physically visit offices to submit architectural layouts, while payment verification requires municipal staff to visually check SMS notifications on personal mobile devices. This reliance on manual reconciliation creates operational processing bottlenecks and property record vulnerabilities.

Concurrently, Somalia features an exceptionally high adoption of mobile financial services, serving as the main financial ecosystem for daily commercial transactions. However, a distinct operational disconnect exists between this widespread digital finance adoption and municipal workflows, as local permitting tools lack direct automated gateway integration.

To bridge this gap between urban growth, private digital finance, and municipal capacity, this study proposes the design and implementation of an Integrated Web and Mobile-Based Building Permit System for the Banadir Regional Administration. The system links a centralized Web Platform for staff review with a synchronized Mobile Application for property owners and field inspectors. By integrating dynamic digital verification tools alongside automated payment gateway capabilities, this study provides a secure, efficient, and enforceable digital framework for metropolitan construction oversight.

1.2 Problem Statement

In modern municipal administration, an integrated digital permit system is essential for maintaining urban spatial order, ensuring structural safety compliance, and streamlining public service delivery. Ideally, a metropolitan municipality operates a synchronized web and mobile building permit infrastructure where property developers and citizens submit construction requests remotely, track review progress transparently, and complete mandatory permit fees through integrated digital payment gateways (Fauth et al., 2026). Under an optimal governance framework, municipal staff review submitted application files from a centralized dashboard, while field inspectors utilize mobile verification tools such as dynamic digital verification scanning and permit validation to cross-reference approved construction limits directly at physical building sites to prevent unauthorized structural modifications.

However, the operational reality within the district offices of the Banadir Regional Administration (BRA) presents a significant contrast to these digital governance standards. Local municipal authorities rely on isolated counter-entry software that lacks external applicant accessibility, resulting in severe administrative fragmentation (Amer et al., 2025).

Property owners are forced to physically visit municipal offices and stand in queues to hand over physical architectural layouts for manual counter entry. Financially, the absence of an integrated payment gateway forces citizens to perform manual mobile transfers at counters, where staff visually check transaction notifications on handheld screens and manually type reference details into local logs.

This operational fragmentation creates severe administrative bottlenecks, revenue leakages, unrecorded financial entries, and cumbersome manual reconciliation routines. Crucially, municipal field inspectors lack real-time mobile access to approved building permit records while conducting field visits. As a result, property developers can easily bypass authorized construction limits such as constructing a multi-story building on a plot approved strictly for a low density residential structure without immediate detection or automated verification by local authorities. Furthermore, requiring physical office visits and manual payment checks creates institutional friction, erodes public trust, and leaves sensitive property records vulnerable to data corruption.

Without transitioning to an integrated, dual-platform digital solution, municipal permitting operations will continue to struggle with administrative inefficiencies, unverified fee collection, and compromised regulatory control over rapid urban growth. Therefore, this study directly addresses these operational vulnerabilities by engineering an Integrated Web and Mobile-Based Building Permit System tailored for the Banadir Regional Administration. Unlike internal counter-entry software, the proposed system enables property owners to securely apply, upload blueprints, track progress, and complete payments remotely. Furthermore, it establishes superior data integrity for property records while equipping field

inspectors with dynamic mobile scanning capabilities to ensure strict alignment between approved digital permits and physical site developments.

1.3 Research Questions

The following research questions will be addressed during this study:

1. What are the key operational inefficiencies, manual payment verification gaps, and compliance enforcement challenges in the existing building permit process within the Banadir Regional Administration?
2. How can a centralized Web Staff Portal be designed and developed to automate permit application handling, architectural blueprint reviews, and administrative approval workflows?
3. How can an automated mobile payment gateway (EVC-Plus API) be seamlessly integrated into the building permit platform to ensure real-time fee processing and automated transaction logging?
4. How can a Mobile Application equipped with QR Code and Permit ID scanning mechanisms be implemented to enable real-time tracking for applicants and empower field inspectors in enforcing site compliance?

1.4 Research Objectives:

1.4.1 General Objective

The primary objective of this study is to design, develop, and test an Integrated Web and Mobile-Based Building Permit System featuring automated EVC-Plus payment integration and mobile QR Code/Permit ID verification tools to streamline permitting operations and enhance site compliance for the Banadir Regional Administration.

1.4.2 Specific Objectives

To achieve the general objective, the study will address the following four implementable objectives:

- To evaluate the existing building permit workflow within municipal offices to identify functional processing bottlenecks, manual payment verification vulnerabilities, and field compliance gaps.
- To design and implement a centralized Web Staff Portal for municipal personnel to receive, review, and process building permit applications along with architectural documentation.
- To integrate an automated EVC-Plus payment gateway API into the platform to facilitate secure, real-time fee collection and automated transaction reconciliation.
- To develop and validate a synchronized Mobile Application featuring real-time application tracking for property owners and QR Code/Permit ID scanning tools for field inspectors to verify approved construction limits on-site.

1.5 Motivation of the Study

The motivation to undertake this research stems from a strong academic and technical interest in leveraging modern web and mobile software architectures to resolve persistent operational inefficiencies in public governance. Observing the rapid metropolitan expansion in Mogadishu highlights a stark operational contrast between physical urban development and the technological tools utilized by municipal authorities.

Practically, witnessing property developers endure cumbersome counter-based application routines and manual short-message payment reconciliations within municipal offices

emphasizes the urgent need for software automation. Furthermore, while mobile financial ecosystems process the majority of daily civic and commercial transactions, local government administrative workflows remain fundamentally disconnected from direct, automated payment gateway interfaces.

From a software engineering and societal standpoint, this study is driven by the opportunity to engineer an integrated digital solution that bridges this operational gap. Designing a synchronized platform combining a centralized Web Staff Portal with a Mobile Application featuring dynamic verification scanning and real-time payment gateway integration presents an impactful system development challenge. Ultimately, the motivation lies in establishing an enforceable, transparent, and accessible digital framework that modernizes municipal permit processing, reduces administrative friction, and enhances service delivery for the Banadir Regional Administration.

1.6 Significance of the Study

This study holds significant practical, institutional, and technical relevance by bridging the operational gap between municipal administrative governance and mobile financial technology in Mogadishu. By engineering an integrated digital permit framework, this research provides tangible benefits to multiple key stakeholders:

- For the Banadir Regional Administration (BRA): The study provides municipal authorities with a centralized digital platform to automate application processing, eliminate unverified counter transaction risks, and streamline administrative workflows. Transitioning to a centralized web platform reduces operational

processing bottlenecks, minimizes revenue leakages, and enhances data integrity for municipal property records.

- For Property Developers and Citizens: Applicants, including local property owners and diaspora investors, gain a convenient digital channel to submit building permit requests, upload architectural blueprints, process permit fees via an integrated automated payment gateway, and track application progress remotely without requiring physical office visits or third-party intermediaries.
- For Municipal Field Inspectors: The proposed mobile application equips field personnel with real-time dynamic verification and permits identification scanning tools directly at construction sites. This capability enables immediate on-site verification of approved structural limits, thereby curbing unauthorized construction expansions and improving municipal regulatory enforcement.
- For Future Researchers and Software Engineers: This project serves as an applied technical blueprint and practical reference for integrating automated mobile payment APIs and mobile verification mechanisms within public sector e-governance platforms in developing metropolitan contexts

1.7 Scope of the Study

This study focuses on the design, development, and operational testing of an Integrated Web and Mobile-Based Building Permit System tailored specifically for the Banadir Regional Administration in Mogadishu, Somalia. The functional context encompasses the engineering of a centralized Web Staff Portal designed for municipal personnel to receive, evaluate, and process building permit applications alongside architectural blueprints. To support district-level operations, the web platform incorporates multi-tiered role-based access control where

administrative staff are explicitly assigned to specific municipal districts within Mogadishu to handle localized workflows. Furthermore, the platform features an automated fee calculation logic that determines permit fees upon data entry based on structural dimensions (such as area in square meters and number of floors), eliminating manual price entry by the applicant.

On the client and enforcement interfaces, the functional boundary includes a synchronized Mobile Application serving property owners and municipal field inspectors. The mobile application enables applicants to submit documentation, track review stages remotely, and process permit fees through an integrated automated payment API workflow. For site monitoring, the mobile platform arms municipal field inspectors with dynamic verification scanning and permit identification tools to cross-reference approved architectural limits directly at physical construction sites.

Geographically, the boundary of this research is strictly confined to the urban administrative jurisdiction of Mogadishu under the management of the Banadir Regional Administration. Temporally, the execution of this study encompasses the complete Software Development Life Cycle (SDLC) including system analysis, architectural design, prototype implementation, API integration, and user acceptance testing conducted within the timeframe spanning from January 2026 to August 2026.

To maintain project feasibility, the scope is strictly limited to building permit authorization workflows and does not extend to national land titling registries, municipal real estate property taxation, or cross-regional zoning networks. Additionally, the mobile payment module utilizes an API-driven transaction integration tailored for academic demonstration and functional system validation rather than commercial banking deployment.

1.8 Organization of the Study

This project report is structured into five logically connected chapters that systematically guide the reader from the theoretical foundation and system design to the practical implementation and validation of the proposed platform. The structure is organized as follows:

Chapter One: Introduction: This chapter establishes the foundation of the research by presenting the background of the study, the problem statement regarding current municipal permit processes, the primary research questions, and the specific implementable objectives. It also outlines the study's motivation, significance to key stakeholders, operational scope, and overall structural organization.

Chapter Two: Literature Review: This chapter examines existing scholarly work, global e-government frameworks, and current municipal permitting systems. It explores theoretical concepts surrounding web-based administrative portals, mobile application technology, and automated mobile payment integration, thereby establishing the conceptual framework necessary to inform the technical architecture of the platform.

Chapter Three: System Methodology and Design: This chapter details the software engineering methodology utilized to address the research objectives. It encompasses detailed system requirements analysis, database schema modeling, dynamic fee calculation algorithms, and high-level architectural design for both the Web Staff Portal and the Mobile Application.

Chapter Four: System Implementation, Testing, and Results: This chapter focuses on the execution and functional validation of the developed platform. It presents the actual user interface implementations, automated EVC-Plus API payment flows, and mobile QR

Code/Permit ID field verification workflows. Furthermore, it details the system testing procedures, performance evaluations, and user feedback collected to confirm that the platform effectively fulfills the study's objectives.

Chapter Five: Summary, Conclusion, and Recommendations: This final chapter synthesizes the primary findings of the project, evaluates how successfully the research objectives were met, discusses system limitations, and offers strategic recommendations for future enhancements and broader municipal deployments.

CHAPTER II: LITERATURE REVIEW

2.1 Introduction

A building permit is an official authorization issued by a government authority that allows individuals or organizations to construct, renovate, extend, or demolish buildings in accordance with established planning regulations, building codes, and safety standards. Building permit systems play a vital role in urban development by ensuring that construction activities comply with legal, environmental, and structural requirements. In many countries, municipalities are responsible for managing the building permit process to promote orderly urban growth, protect public safety, and ensure sustainable land use.

Traditionally, building permit administration has relied on paper-based procedures that require applicants to visit municipal offices to obtain application forms, submit required documents, pay permit fees, and collect approved permits. Although these manual processes have served municipalities for many years, they are often associated with lengthy processing times, inefficient document management, limited transparency, and increased operational costs. Paper-based systems also make it difficult for citizens to monitor application progress and for municipalities to maintain accurate records of permit transactions (Heeks, 2002).

Recent advances in Information and Communication Technology (ICT) have enabled governments worldwide to modernize public service delivery through digital transformation initiatives. Electronic government (e-government), mobile government (m-government), cloud computing, and digital public infrastructure have significantly improved the efficiency, accessibility, and transparency of public services. These technologies enable governments to automate administrative processes, improve communication with citizens, reduce paperwork, and strengthen accountability within public institutions (OECD, 2020; United Nations, 2022).

Digital transformation has become one of the primary strategies adopted by governments to improve public sector performance. Rather than simply converting paper records into electronic documents, digital transformation involves redesigning organizational processes through the effective use of digital technologies to deliver more efficient, transparent, and citizen-centered services (Vial, 2019). Within municipal administrations, digital transformation has supported the development of online service platforms that allow citizens to submit applications, make electronic payments, receive notifications, and monitor service requests remotely.

Building permit management has also benefited from these technological developments. Many municipalities now utilize web-based permit systems that enable online application submission, electronic document management, workflow automation, and digital communication between applicants and government officials. Some advanced municipalities have further integrated mobile applications, Geographic Information Systems (GIS), Building Information Modeling (BIM), and cloud-based services into their permit administration processes to improve decision-making and service delivery (Eastman et al., 2011; Noardo et al., 2020).

Despite these global developments, many municipalities in developing countries continue to experience challenges in delivering efficient building permit services. These challenges include limited digital infrastructure, dependence on manual administrative procedures, centralized decision-making, inadequate service accessibility, and limited opportunities for citizens to interact directly with digital government systems (Heeks, 2003). Consequently, many governments continue to explore practical digital solutions that improve municipal services while remaining compatible with local institutional and technological conditions.

Mogadishu Municipality has initiated the digitalization of its building permit management process by implementing a computerized permit management system. Compared with the previous paper-based approach, the existing system has improved electronic record management and reduced several manual administrative activities. However, field investigation conducted during this research indicates that several operational limitations remain. Citizens generally visit municipal offices to initiate new permit applications, permit fees are collected during application submission with paper receipts serving as proof of payment, and application approval depends on a single authorized officer, which may delay service delivery when that officer is unavailable. In addition, applicants have limited access to digital self-service functions such as online application submission, real-time status tracking, and automated notifications.

To address these operational challenges, this research proposes the design and development of a Web and Mobile Digital Building Permit System for Mogadishu Municipality. The proposed system enables citizens to submit permit applications remotely through a mobile application while preserving office-assisted services for applicants who require support from municipal staff. The system further introduces digital payment records, electronic receipts, real-time application tracking, automated notifications, QR code-based permit verification,

centralized document management, and role-based access control to improve service efficiency, transparency, and accessibility.

This chapter reviews the literature relevant to digital municipal building permit systems and establishes the theoretical and conceptual foundation for the study. Specifically, it discusses the evolution of building permit management systems, digital government technologies, mobile government services, and Digital Public Infrastructure (DPI). The chapter further examines the Technology Acceptance Model (TAM), the Unified Theory of Acceptance and Use of Technology (UTAUT), and Digital Transformation Theory, reviews related empirical studies, presents a comparative analysis of existing approaches, identifies the research gap, and provides the theoretical justification for the proposed Web and Mobile Digital Building Permit System.

2.2 Overview of Digital Municipal Permitting Systems

Municipal governments are responsible for providing essential public services that support urban planning, infrastructure development, environmental protection, and public safety. Among these responsibilities, building permit administration is one of the most critical regulatory functions because it ensures that construction projects comply with planning regulations, building standards, and legal requirements before development begins. As cities continue to grow and urbanization increases, municipalities face greater pressure to deliver permit services efficiently while maintaining transparency, accountability, and regulatory compliance.

Digital municipal permitting systems have emerged as an effective solution for improving the efficiency of building permit administration. These systems integrate information and communication technologies into municipal workflows to automate application processing,

document management, fee collection, approval procedures, and communication between citizens and government officials. Compared with traditional paper-based systems, digital permitting systems reduce administrative workload, improve record management, shorten processing times, and provide citizens with more accessible and transparent public services (United Nations, 2022).

The evolution of municipal permitting systems reflects the broader transformation of digital government. Earlier systems focused primarily on digitizing paper records and supporting internal administrative functions. More recent systems emphasize citizen-centered service delivery by enabling online applications, electronic payments, mobile access, automated notifications, and real-time tracking of service requests. These improvements contribute to greater public satisfaction while enabling municipalities to manage increasing service demands more effectively (OECD, 2020).

The following sections discuss the evolution of building permit systems, beginning with traditional paper-based approaches, existing digital permit systems, and the proposed Web and Mobile Digital Building Permit System developed for this research.

2.2.1 Building Permit Systems

A building permit system is an administrative mechanism established by government authorities to regulate construction activities and ensure that all building projects comply with applicable planning regulations, zoning requirements, building codes, environmental standards, and public safety regulations. Building permits provide legal authorization for construction activities and enable municipalities to monitor urban development while protecting public health, safety, and the environment (Eastman et al., 2011).

Municipal governments use building permit systems to evaluate proposed construction projects before work begins. During the permit process, officials review architectural drawings, structural designs, land ownership documents, and other supporting information to determine whether a proposed development satisfies legal and technical requirements. Only after the application has successfully passed the review process is a building permit issued, allowing construction to proceed legally (Noardo et al., 2020).

Beyond regulatory compliance, building permit systems contribute significantly to urban planning and sustainable development. They assist municipalities in controlling land use, preventing unauthorized construction, ensuring infrastructure compatibility, and maintaining accurate records of urban development. Building permit records also support future planning decisions by providing reliable information on construction activities within a municipality (United Nations, 2022).

With the advancement of Information and Communication Technology (ICT), building permit administration has evolved considerably. Earlier permit systems relied almost entirely on manual procedures and paper documentation, whereas modern systems increasingly utilize web-based platforms, cloud computing, digital databases, and mobile technologies to improve efficiency and accessibility. These technological developments have enabled municipalities to automate administrative tasks, reduce paperwork, improve service transparency, and strengthen communication between government agencies and citizens (OECD, 2020).

Despite these developments, the degree of digital transformation differs considerably among municipalities. While some local governments provide fully integrated online permitting services, others continue to rely on partially digital systems that retain several manual

administrative procedures. The level of digital maturity often depends on factors such as technological infrastructure, financial resources, organizational readiness, and government policies supporting digital transformation (Vial, 2019).

2.2.2 Traditional Paper-Based Building Permit Systems

Before the widespread adoption of digital government technologies, most municipalities managed building permits through paper-based administrative procedures. Under this approach, applicants were required to visit municipal offices physically to obtain application forms, submit supporting documents, pay permit fees, and collect approved permits. Government officers manually recorded application information, verified submitted documents, coordinated reviews among departments, and stored completed records in physical filing systems (Heeks, 2002).

The traditional paper-based process typically consists of several sequential stages. Applicants first prepare the required documents, including land ownership records, architectural drawings, engineering plans, and identification documents. These documents are submitted physically to the responsible municipal office, where administrative staff review the completeness of the application before forwarding it to technical departments for evaluation. Following inspection and approval, applicants are notified to collect the permit in person (Gil-Garcia, 2012).

Although paper-based systems require relatively little technological infrastructure, they present several operational challenges. Manual processing often increases the time required to review applications because documents must move physically between departments. Paper records are also vulnerable to loss, damage, duplication, and unauthorized alteration, making long-term record management difficult. Furthermore, citizens generally have limited

visibility into the status of their applications, requiring repeated visits or telephone inquiries to obtain updates (Heeks, 2003).

Another limitation of traditional systems is the significant administrative workload placed on municipal staff. Employees spend considerable time handling paper documents, recording information manually, searching archived files, and preparing official correspondence. As the number of permit applications increases, these activities may reduce organizational efficiency and delay public service delivery (United Nations, 2022).

Despite these limitations, paper-based systems continue to be used in many developing countries because they require relatively low investment in technological infrastructure and are familiar to both municipal employees and citizens. However, increasing urbanization and rising public expectations for efficient government services have encouraged many municipalities to adopt digital permit management systems that reduce dependence on manual administrative procedures (OECD, 2020).

2.2.3 Existing Digital Building Permit Systems

The increasing adoption of digital government has encouraged municipalities worldwide to replace or improve traditional paper-based permit administration with computerized systems. Digital building permit systems integrate databases, electronic document management, workflow automation, and online communication to improve administrative efficiency while maintaining compliance with planning regulations and building standards (United Nations, 2022).

Modern digital permit systems commonly support electronic application management, centralized document storage, digital workflow management, and automated communication among municipal departments. Applicants may submit supporting documents electronically,

while authorized officers review applications through integrated information systems rather than relying solely on physical documentation. These improvements reduce paperwork, improve record accuracy, and enhance coordination between administrative units (Gil-Garcia, 2012).

However, digital transformation is not always complete. Many municipalities implement partially digital systems, where some administrative activities remain manual despite the use of computerized databases. Examples include physical submission of applications, manual payment verification, paper receipts, or centralized approval procedures that continue to depend heavily on individual officials. Consequently, although information management improves, citizens may still experience delays and limited access to digital self-service functions (Vial, 2019).

Existing Building Permit System in Mogadishu Municipality

The field investigation conducted for this study indicates that Mogadishu Municipality currently operates a computerized building permit management system that has significantly improved electronic record management compared with the previous paper-based process. The system allows municipal staff to record permit information digitally and supports computerized management of permit records.

However, the investigation also identified several operational characteristics that continue to affect service delivery:

- Applicants generally visit municipal offices to initiate new building permit applications.
- Municipal staff submit applications into the digital system on behalf of applicants.

- Permit fees are collected during application submission.
- Applicants receive a paper receipt as proof of payment.
- Applications require approval by one authorized officer before permits are issued.
- When the authorized approver is unavailable, application processing may be delayed.
- After approval, payment records are transferred through internal municipal procedures.
- Citizens have limited opportunities to submit applications independently, monitor application progress in real time, or receive automated electronic notifications.

These findings indicate that while the municipality has made meaningful progress toward digital transformation, several important interactions between citizens and the permit system continue to rely on physical visits and manual administrative activities. These operational observations provide the practical foundation for the proposed improvements presented in this study.

2.2.4 Proposed Web and Mobile Digital Building Permit System

Based on the operational challenges identified through the literature review and field investigation, this study proposes a Web and Mobile Digital Building Permit System for Mogadishu Municipality. The proposed system is designed to improve service accessibility, administrative efficiency, transparency, and citizen participation while remaining compatible with existing municipal procedures.

Unlike the existing process, the proposed system allows citizens to submit building permit applications directly through a mobile application without visiting municipal offices. At the same time, applicants who require assistance may continue to visit the municipal office,

where authorized staff can submit applications on their behalf. This hybrid approach ensures that digital services remain inclusive for citizens with different levels of digital literacy.

The proposed system integrates digital payment recording during application submission, enabling payment information to be stored electronically while generating an electronic receipt that serves as proof of payment. This approach reduces dependence on manual paper receipts and improves financial record management. To improve administrative efficiency, the system requires only one approval before a permit is issued.

Approval may be performed by either the system administrator or an authorized municipal staff member, depending on organizational responsibilities. This flexible authorization mechanism reduces delays that may occur when a single official is unavailable while maintaining appropriate administrative control.

Additional features of the proposed system include real-time application status tracking, automated notifications, centralized digital document management, role-based access control, and QR code-based permit verification. Together, these features improve communication between citizens and the municipality, strengthen transparency, simplify permit verification, and support more efficient management of building permit services.

The proposed system therefore represents an enhancement of the municipality's current digital permit process rather than a complete replacement. By combining mobile accessibility with existing office-assisted services, the system supports the ongoing digital transformation of municipal building permit administration while addressing the practical operational challenges identified during the field investigation.

2.2.5 Web Government (E-Government)

Electronic Government (e-Government), commonly referred to as Web Government, is the use of Information and Communication Technologies (ICT), particularly internet-based technologies, by government institutions to improve the delivery of public services, strengthen administrative efficiency, and promote transparency and accountability. The Organization for Economic Co-operation and Development (OECD) defines digital government as the strategic use of digital technologies and data to create public value by transforming the way governments design, deliver, and evaluate public services (OECD, 2020).

The concept of e-Government extends beyond simply digitizing existing paper-based services. It involves redesigning government processes to improve communication, collaboration, and service delivery among government institutions, businesses, and citizens. According to the (United Nations, 2022), digital government enables governments to become more responsive, citizen-centered, transparent, and inclusive while supporting sustainable development and improving public trust.

E-Government services are commonly categorized into four interaction models:

- Government-to-Citizen (G2C): Digital services provided directly to citizens, such as online applications, tax payment, permit applications, and public information services.
- Government-to-Business (G2B): Digital interactions between government agencies and businesses, including licensing, procurement, and regulatory compliance.
- Government-to-Government (G2G): Information sharing and collaboration among government departments and agencies.

- Government-to-Employee (G2E): Digital systems that support government employees in administrative and operational activities (Gil-Garcia, 2012).

Among these interaction models, Government-to-Citizen (G2C) services are particularly relevant to building permit administration because they allow citizens to access public services without repeatedly visiting government offices. Online permit application systems enable applicants to submit documents, monitor application progress, receive notifications, and communicate with municipal officials electronically, thereby improving both efficiency and citizen satisfaction (United Nations, 2022).

The implementation of e-Government also provides significant organizational benefits for municipalities. Digital systems improve document management, reduce paperwork, facilitate information sharing among departments, and minimize processing delays caused by manual administrative procedures. Furthermore, centralized databases improve the accuracy, security, and accessibility of government records while reducing the risks associated with physical document storage (OECD, 2020).

Despite these advantages, implementing e-Government in developing countries presents several challenges. According to (Heeks, 2003), limited ICT infrastructure, insufficient financial resources, organizational resistance to change, inadequate technical skills, and low digital literacy among citizens remain significant barriers to successful digital government implementation. Governments must therefore design digital services that reflect local technological capabilities and user needs rather than simply adopting solutions developed for technologically advanced environments.

For municipal building permit services, e-Government provides the technological foundation for replacing manual administrative processes with integrated digital workflows. By enabling

electronic application management, digital document processing, and online communication, municipalities can improve service accessibility while maintaining regulatory compliance and administrative control. These characteristics directly support the objectives of the proposed Web and Mobile Digital Building Permit System developed in this research.

2.2.6 Mobile Government (M-Government)

Mobile Government (m-Government) refers to the delivery of government information and public services through mobile technologies such as smartphones, tablets, and other wireless devices. It extends the capabilities of e-Government by allowing citizens to access government services regardless of their location, thereby increasing convenience, accessibility, and service availability (United Nations, 2022).

The rapid growth of smartphone usage has significantly influenced the development of mobile government services. Compared with traditional web-based systems that often require desktop or laptop computers, mobile applications provide more flexible and convenient access to public services. Citizens can submit applications, upload documents, receive notifications, and communicate with government agencies directly from their mobile devices without visiting government offices.

Mobile government offers several advantages over conventional electronic government systems. First, it increases accessibility because smartphones are widely used across both developed and developing countries. Second, mobile applications support real-time communication through push notifications, allowing governments to inform citizens immediately about application progress, approval decisions, or additional document requirements. Third, mobile devices integrate features such as cameras, GPS, QR code

scanning, and biometric authentication, enabling governments to deliver more secure and interactive digital services.

In the context of building permit administration, mobile government enables applicants to submit permit requests remotely, upload supporting documents using smartphone cameras, monitor application status in real time, and receive electronic notifications throughout the approval process. These capabilities reduce unnecessary travel to municipal offices while improving communication between citizens and government officials.

Although mobile government offers substantial benefits, successful implementation depends on several factors, including reliable internet connectivity, smartphone availability, user digital literacy, and secure application design. Governments must therefore ensure that mobile services remain simple, secure, and inclusive so that citizens with varying technological experience can effectively use digital public services (Al-Hujran et al., 2015).

The proposed system developed in this research adopts a mobile government approach by providing a dedicated mobile application through which citizens can submit building permit applications, monitor application progress, receive notifications, access digital receipts, and verify approved permits using QR codes.

At the same time, the system preserves office-assisted application services for citizens who require support from municipal staff, ensuring that digital transformation remains inclusive rather than exclusive.

2.2.7 Digital Public Infrastructure (DPI)

Digital Public Infrastructure (DPI) refers to the foundational digital systems that enable governments, businesses, and citizens to securely access digital services. According to the United Nations Development Programme (UNDP, 2024), DPI consists of interoperable

digital platforms that provide shared public infrastructure for digital identity, digital payments, and secure data exchange, enabling governments to deliver efficient, inclusive, and citizen-centered public services.

Unlike standalone software applications, Digital Public Infrastructure provides reusable digital building blocks that can support multiple government services simultaneously. Rather than developing isolated systems for individual government departments, DPI encourages integration between institutions through standardized digital platforms and secure information sharing.

Three core components commonly characterize Digital Public Infrastructure:

1. Digital Identity, which enables secure authentication of citizens and government officials when accessing public services.
2. Digital Payment Systems, which facilitate secure electronic financial transactions between citizens and government institutions.
3. Data Exchange Platforms, which enable authorized government agencies to securely share and verify information while maintaining data privacy and integrity (WorldBank, 2020).

For municipal governments, DPI provides the technological foundation for integrating permit management with other public services. Digital payment infrastructure supports electronic collection and recording of permit fees, while secure databases improve record management and reduce duplication of information. Data-sharing mechanisms also enable government departments to coordinate more efficiently during permit evaluation and approval.

Although comprehensive Digital Public Infrastructure continues to evolve globally, its principles are increasingly influencing the design of municipal information systems. Governments are gradually adopting interoperable digital platforms that improve service efficiency while reducing administrative complexity and operational costs (United Nations, 2022).

The proposed Web and Mobile Digital Building Permit System incorporates several concepts associated with Digital Public Infrastructure. These include electronic payment recording, centralized digital databases, secure user authentication, QR code-based permit verification, and integrated digital document management.

While the proposed system does not represent a complete national DPI platform, it applies fundamental DPI principles to improve the efficiency, transparency, and accessibility of municipal building permit administration in Mogadishu Municipality.

2.3 Theoretical Framework

A theoretical framework provides the conceptual foundation for a research study by explaining the theories that support the research problem, objectives, and proposed solutions. It guides researchers in understanding the relationships between variables and provides a basis for interpreting research findings. In information systems research, theoretical frameworks are widely used to explain technology adoption, user behavior, organizational change, and digital transformation.

This study adopts three complementary theories: the Technology Acceptance Model (TAM), the Unified Theory of Acceptance and Use of Technology (UTAUT), and Digital Transformation Theory. Together, these theories explain the factors influencing user

acceptance of digital systems, organizational adoption of technology, and the transformation of public service delivery through digital technologies.

2.3.1 Technology Acceptance Model (TAM)

The Technology Acceptance Model (TAM) was developed by Fred D. Davis in 1989 to explain why individuals accept or reject new information technologies. TAM is one of the most influential and widely applied theories in Information Systems research because it provides a simple but effective explanation of user behavior toward technology adoption (Davis, 1989).

According to TAM, two primary factors determine whether users are willing to adopt a new information system:

Perceived Usefulness (PU)

Perceived usefulness refers to the degree to which a person believes that using a particular system will improve his or her job performance or daily activities. When users believe that a system helps them accomplish tasks more efficiently, they are more likely to accept and use it.

Within the proposed Web and Mobile Digital Building Permit System, perceived usefulness is reflected through features such as:

- Mobile submission of permit applications.
- Real-time application tracking.
- Electronic payment records.
- Digital receipts.

- QR code permit verification.
- Faster communication between applicants and municipal officers.

These features reduce unnecessary office visits and improve the efficiency of permit processing for both citizens and municipal employees.

Perceived Ease of Use (PEOU)

Perceived ease of use refers to the degree to which a person believes that a system can be used with minimal effort. Systems that are simple, intuitive, and easy to learn are more likely to be accepted by users.

For the proposed system, ease of use is supported through:

- User-friendly mobile interfaces.
- Simple application forms.
- Clear navigation.
- Automated notifications.
- Minimal steps required to complete permit applications.

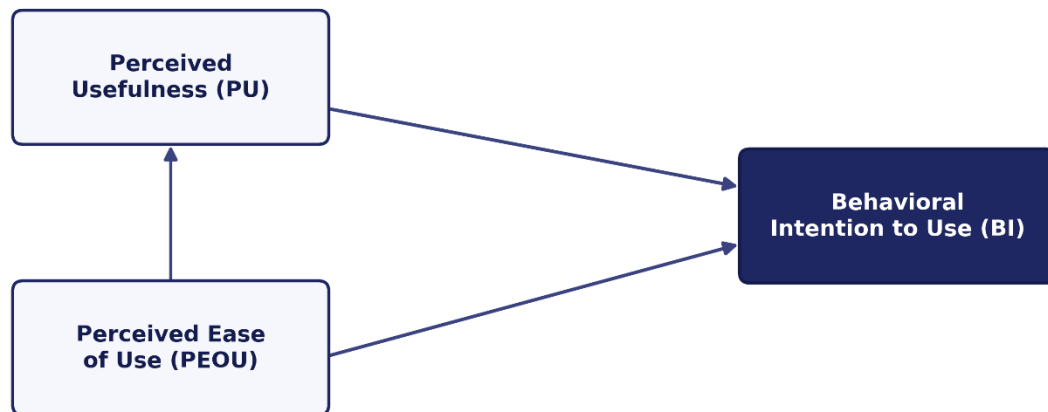
Maintaining office-assisted application services further supports ease of use by allowing citizens who are less familiar with digital technologies to receive assistance from municipal staff.

Relevance of TAM to This Study

TAM provides an appropriate theoretical foundation because the success of the proposed system depends largely on whether citizens and municipal employees are willing to adopt and continuously use the digital platform. If users perceive the system as useful and easy to

operate, they are more likely to transition from traditional manual procedures to digital permit services.

Figure 2.1 Technology Acceptance Model (TAM)



Adapted from (Davis, 1989)

2.3.2 Unified Theory of Acceptance and Use of Technology (UTAUT)

The Unified Theory of Acceptance and Use of Technology (UTAUT) was proposed by (Venkatesh et al., 2003). The theory integrates concepts from eight earlier technology acceptance models into a unified framework capable of explaining user intention and actual technology usage more comprehensively than previous models.

UTAUT identifies four major determinants of technology adoption.

Performance Expectancy

Performance expectancy refers to the degree to which users believe that using a system will improve their performance.

For the proposed building permit system, users are expected to benefit from:

- Faster permit processing.
- Improved communication.
- Reduced travel to municipal offices.
- Better access to permit information.

Effort Expectancy

Effort expectancy refers to the ease associated with using a technology.

The proposed system emphasizes:

- Simple mobile interfaces.
- Easy document uploading.
- Clear application procedures.
- Straightforward navigation.

These characteristics encourage citizens with different levels of digital literacy to use the system effectively.

Social Influence

Social influence describes the extent to which individuals believe that important people or organizations encourage the use of technology.

Within municipal services, social influence may arise from:

- Municipal awareness campaigns.

- Recommendations from municipal staff.
- Community encouragement.
- Government policies promoting digital services.

Facilitating Conditions

Facilitating conditions refer to the availability of organizational and technical resources necessary to support technology use.

Examples include:

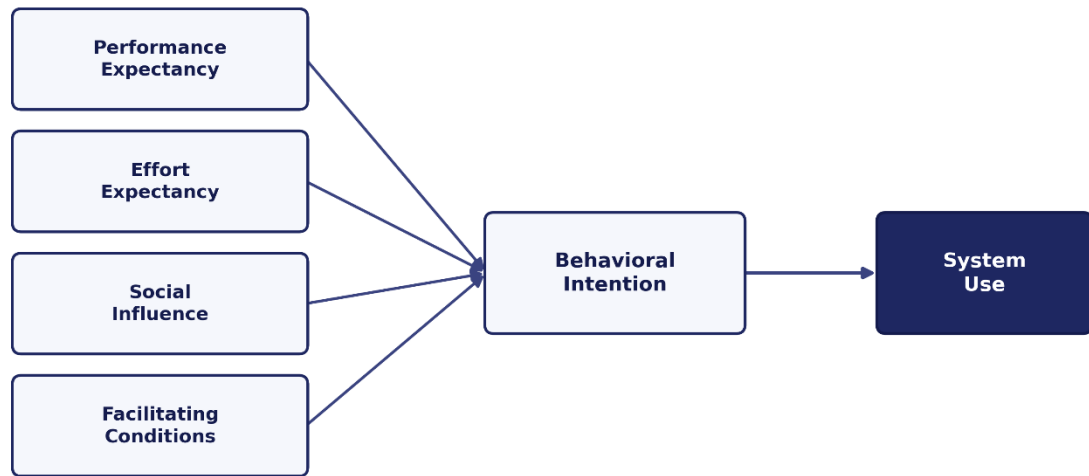
- Smartphone availability.
- Internet connectivity.
- Technical support.
- Staff training.
- Reliable municipal ICT infrastructure.

Providing both mobile self-service and office-assisted services strengthens facilitating conditions by ensuring that all citizens can access the permit system regardless of their technological experience.

Relevance of UTAUT to This Study

UTAUT is particularly suitable because it considers both individual and organizational factors influencing technology adoption. Unlike TAM, which focuses mainly on usefulness and ease of use, UTAUT incorporates infrastructure, organizational support, and social influence, making it highly applicable to public sector digital transformation.

Figure 2.2 Unified Theory of Acceptance and Use of Technology (UTAUT)



Adapted from (Venkatesh et al., 2003).

2.3.3 Digital Transformation Theory

Digital Transformation Theory explains how organizations use digital technologies to redesign business processes, improve operational performance, and create greater value for users. Unlike simple digitization, which involves converting paper records into electronic form, digital transformation requires fundamental changes in organizational processes, service delivery, and decision-making through the strategic use of digital technologies (Vial, 2019).

In the public sector, digital transformation aims to improve government services by increasing efficiency, transparency, accountability, and citizen participation. Governments increasingly adopt digital technologies not only to automate administrative tasks but also to redesign service delivery models that better meet citizens' expectations (Mergel et al., 2019).

Digital transformation typically involves several key components:

- Digital service delivery.
- Process automation.
- Data-driven decision making.
- Improved citizen engagement.
- Organizational innovation.
- Continuous technological improvement.

For municipal governments, digital transformation enables traditional administrative procedures to evolve into integrated digital services that reduce paperwork, improve communication, and enhance operational efficiency.

Relevance to This Study

The proposed Web and Mobile Digital Building Permit System directly supports digital transformation within Mogadishu Municipality by redesigning the existing permit process rather than merely computerizing individual tasks. The system introduces:

- Mobile application submission.
- Digital payment recording.
- Electronic receipts.
- Automated notifications.
- QR code verification.
- Centralized digital records.

- Real-time application tracking.

These improvements contribute to greater efficiency, transparency, and citizen-centered service delivery while maintaining compatibility with existing municipal operations.

Table 2.1 Comparison of Theoretical Frameworks

Theory	Main Focus	Key Constructs	Contribution to This Study
Technology Acceptance Model (TAM)	Individual technology acceptance	Perceived Usefulness, Perceived Ease of Use	Explains citizen and staff acceptance of the proposed permit system.
Unified Theory of Acceptance and Use of Technology (UTAUT)	Technology adoption in organizations	Performance Expectancy, Effort Expectancy, Social Influence, Facilitating Conditions	Explains factors influencing adoption and continued use of the system.
Digital Transformation Theory	Organizational transformation through digital technologies	Process redesign, digital services, organizational change	Provides the conceptual foundation for redesigning the municipal permit process into a citizen-centered digital service.

Together, these theories provide a comprehensive conceptual foundation for designing, implementing, and evaluating the proposed Web and Mobile Digital Building Permit System for Mogadishu Municipality. TAM explains how users perceive and accept the system, UTAUT broadens this perspective by considering organizational and environmental factors affecting technology adoption, and Digital Transformation Theory frames the redesign of municipal service delivery through digital technologies. Collectively, they support both the technological and organizational objectives of this study.

2.4 Related Work

The rapid advancement of digital technologies has significantly transformed the delivery of government services worldwide. Governments are increasingly adopting electronic government (e-government), mobile government (m-government), and digital transformation strategies to improve service accessibility, operational efficiency, transparency, and citizen engagement. Municipal governments have also embraced these technologies to modernize administrative services, including building permit management. Previous studies have demonstrated that digital permit systems reduce administrative burdens, improve record management, and enhance communication between citizens and government institutions. However, the literature also indicates that many municipalities continue to experience challenges associated with organizational readiness, limited digital infrastructure, manual administrative procedures, and user adoption. This section reviews previous studies according to major research themes relevant to the proposed Web and Mobile Digital Building Permit System.

2.4.1 Digital Transformation in Municipal Government

Digital transformation has become one of the primary strategies adopted by governments to improve public administration and public service delivery. Unlike simple digitization, digital

transformation involves redesigning organizational processes, governance structures, and service delivery models through the effective use of digital technologies. According to (Vial, 2019), successful digital transformation requires organizations to fundamentally rethink how services are designed, delivered, and continuously improved to create greater value for users.

Within municipal governments, digital transformation has enabled local authorities to automate administrative workflows, improve document management, strengthen collaboration among departments, and provide citizens with more accessible digital services. Rather than relying solely on traditional paper-based procedures, municipalities increasingly use integrated information systems to support licensing, tax administration, land management, urban planning, and permit administration (Mergel et al., 2019).

The (United Nations, 2022) survey reports that digital transformation has improved government efficiency, transparency, and citizen participation across many countries. Governments that invest in digital public services are better positioned to deliver timely services, reduce administrative costs, and strengthen public trust.

Similarly, the (OECD, 2020) Digital Government Index emphasizes that digital transformation should focus on creating citizen-centered services supported by data-driven decision-making and integrated digital platforms rather than simply converting manual procedures into electronic formats.

Despite these benefits, digital transformation remains challenging, particularly in developing countries. (Heeks, 2003) argues that many digital government initiatives experience implementation difficulties because they fail to consider organizational capacity, institutional practices, and local technological conditions. Challenges such as limited ICT infrastructure,

insufficient technical expertise, financial constraints, and resistance to organizational change often reduce the effectiveness of digital government projects.

These findings suggest that successful municipal digital transformation requires more than adopting new technologies. Digital solutions must also reflect local administrative practices, institutional capacity, and citizen needs. This perspective directly supports the present study, which proposes a Web and Mobile Digital Building Permit System designed around the operational realities identified through field investigation in Mogadishu Municipality rather than adopting a generic digital solution.

2.4.2 Digital Building Permit Management Systems

Building permit administration has experienced significant digital transformation over the past decade as municipalities seek to improve efficiency and reduce reliance on manual administrative procedures. Modern digital building permit systems integrate electronic document management, workflow automation, centralized databases, and online communication to simplify permit processing while maintaining compliance with planning regulations and construction standards.

(Eastman et al., 2011) explain that digital technologies such as Building Information Modeling (BIM) improve information management throughout construction projects by enabling accurate digital representation and efficient exchange of building information. Although BIM was initially developed for construction project management, its principles have increasingly influenced digital permitting processes by improving document quality and facilitating collaboration among stakeholders.

Similarly, (Noardo et al., 2020) examined the integration of Geographic Information Systems (GIS) and Building Information Modeling (GeoBIM) within digital permit management.

Their study found that integrating digital spatial information into permit approval processes improves decision-making, increases information accuracy, and reduces administrative complexity.

The researchers also observed that digital permit systems support more transparent communication between applicants and municipal authorities while improving regulatory compliance.

The literature further indicates that digital building permit systems reduce processing time, improve document security, and facilitate centralized information management. Electronic submission of applications minimizes dependence on physical documentation, while automated workflows improve coordination among departments responsible for permit review and approval. These improvements enhance both administrative efficiency and service quality.

However, previous studies also recognize that implementing advanced digital permitting technologies often requires significant investment in ICT infrastructure, skilled personnel, and institutional capacity. Consequently, municipalities in developing countries frequently adopt partially digital systems that combine computerized information management with manual administrative procedures. This demonstrates that digital transformation is typically an evolutionary process rather than an immediate replacement of traditional systems.

The proposed Web and Mobile Digital Building Permit System follows this practical approach by strengthening existing digital permit management while introducing additional citizen-centered services, including mobile application submission, electronic payment records, digital receipts, QR code verification, and real-time application tracking.

2.4.3 Mobile Government Services

The rapid growth of smartphone technology has significantly influenced the way governments deliver public services. While traditional e-government services primarily relied on web portals accessed through desktop or laptop computers, the increasing availability of mobile devices has enabled governments to provide services that are more accessible, flexible, and convenient for citizens. This evolution has led to the development of Mobile Government (m-Government), which extends electronic government services through mobile applications and wireless communication technologies.

According to the (United Nations, 2022) survey, mobile government has become an important component of digital public service delivery because it enables citizens to access government services regardless of time and location. Mobile applications reduce geographical barriers, improve communication between governments and citizens, and increase the accessibility of public services, particularly in regions where smartphones are more common than personal computers.

Similarly, (Al-Hujran et al., 2015) found that citizens are more willing to adopt mobile government services when applications are perceived as useful, easy to use, secure, and trustworthy. The study emphasizes that user confidence in mobile applications plays a critical role in determining whether citizens continue using digital government services. Governments must therefore design mobile applications that provide simple navigation, reliable system performance, and secure handling of users' personal information.

The (OECD, 2020) also highlights that modern digital governments should provide services through multiple digital channels, including mobile platforms, to improve citizen engagement and ensure inclusive public service delivery. Rather than requiring citizens to

visit government offices repeatedly, mobile technologies allow users to complete administrative processes remotely while receiving immediate feedback and updates regarding their requests.

Previous studies further demonstrate that mobile government improves communication through features such as push notifications, electronic messaging, digital document submission, and real-time status updates.

These capabilities reduce administrative delays by allowing citizens to receive timely information without contacting government offices directly. Consequently, mobile government contributes to greater service transparency, improved citizen satisfaction, and more efficient public administration.

Despite these advantages, researchers acknowledge that successful implementation of mobile government depends on several factors, including smartphone ownership, internet availability, digital literacy, and adequate technical support. Municipal governments must therefore ensure that digital services remain inclusive by providing alternative service channels for citizens who may experience difficulties using mobile applications.

The proposed Web and Mobile Digital Building Permit System adopts these principles by enabling citizens to submit permit applications, upload supporting documents, monitor application progress, receive automated notifications, and access digital payment records through a mobile application. At the same time, applicants who require assistance may continue visiting the municipal office, where authorized staff can submit applications on their behalf. This hybrid service model supports wider accessibility while encouraging the gradual adoption of digital government services.

2.4.4 Technology Adoption in Digital Government

The successful implementation of digital government systems depends not only on technological infrastructure but also on the willingness of citizens and government employees to adopt and continuously use new technologies. Consequently, numerous studies have investigated the factors influencing technology acceptance within public sector information systems. These studies consistently indicate that system success depends on both technical quality and user acceptance.

One of the earliest and most influential models explaining technology adoption is the Technology Acceptance Model (TAM) developed by (Davis, 1989). The model proposes that users are more likely to adopt an information system when they perceive it as useful and easy to use. Since its introduction, TAM has been extensively applied in studies evaluating e-government services, online public service platforms, and mobile government applications. Research findings consistently demonstrate that systems offering simple interfaces, efficient workflows, and practical benefits achieve higher levels of user acceptance.

To provide a broader explanation of technology adoption, (Venkatesh et al., 2003) introduced the Unified Theory of Acceptance and Use of Technology (UTAUT). Unlike TAM, UTAUT incorporates organizational and environmental factors in addition to individual user perceptions.

The theory identifies performance expectancy, effort expectancy, social influence, and facilitating conditions as the primary determinants influencing technology adoption and continued system usage.

Recent digital government studies have applied these theories to evaluate citizen acceptance of online government services. The findings indicate that citizens are more likely to adopt

digital government platforms when they experience faster service delivery, simplified procedures, transparent communication, and reliable technical support. Conversely, complicated user interfaces, poor internet connectivity, limited technical assistance, and lack of trust in digital systems reduce technology adoption.

The literature also emphasizes that technology adoption within government institutions requires organizational commitment alongside citizen acceptance. Government employees must receive adequate training, management support, and technical resources to effectively operate new digital systems. Without these facilitating conditions, even well-designed information systems may fail to achieve their intended objectives.

These findings strongly support the proposed Web and Mobile Digital Building Permit System. The system has been designed to improve user acceptance by providing an intuitive mobile interface, reducing administrative complexity, enabling real-time application tracking, and simplifying communication between citizens and municipal officials. Furthermore, the inclusion of office-assisted application services ensures that citizens who have limited digital experience can continue accessing permit services without exclusion.

2.4.5 Challenges in Existing Digital Permit Systems

Although previous studies demonstrate that digital technologies significantly improve municipal service delivery, existing digital permit systems still face several limitations that affect their effectiveness, particularly in developing countries and rapidly urbanizing environments. The literature indicates that successful digital transformation is not determined only by the availability of technology but also by the ability of systems to address operational processes, institutional structures, financial mechanisms, and local user requirements.

One of the most common challenges identified in existing digital permit systems is the incomplete transformation of manual procedures into digital workflows. Many municipalities adopt computerized systems that improve record storage and administrative management but continue relying on physical visits, manual verification, paper-based evidence, and centralized approval processes.

(Heeks, 2002, 2003) explains that digital government projects in developing countries frequently experience limited impact when new technologies are introduced without redesigning existing organizational processes. As a result, some systems become digital replacements for individual tasks rather than fully integrated service platforms.

Another challenge relates to limited citizen-centered services. Existing e-permitting solutions often focus primarily on internal municipal administration, while providing fewer capabilities for citizens to independently interact with government services. Many systems allow staff members to manage applications electronically but still require applicants to physically visit government offices for submission, payment confirmation, or status inquiries. According to the (United Nations, 2022), modern digital government services should move beyond administrative automation and provide accessible, transparent, and interactive services that place citizens at the center of service delivery.

A further limitation is the lack of mobile integration within many municipal permit systems. Previous research on digital permitting has mainly focused on web-based platforms designed for desktop environments.

While these systems improve administrative efficiency, they may not fully address accessibility challenges in regions where smartphone usage is significantly higher than desktop computer availability. The (ITU, 2023) highlights that mobile technologies have

become the primary access point to digital services in many developing regions. Therefore, mobile-based government applications represent an important approach for improving public service accessibility.

Existing digital permit systems also demonstrate challenges related to payment management and financial transparency. Traditional permit processes often involve manual fee calculation, physical payment confirmation, and paper-based receipts, creating opportunities for delays, inconsistent records, and financial management difficulties. Digital government literature emphasizes that integrated electronic payment systems improve accountability by creating automated transaction records and reducing dependence on manual cash-handling procedures (OECD, 2020).

Another important limitation is approval dependency and administrative bottlenecks. Some permit systems maintain rigid approval workflows where applications depend on specific officials before progressing to the next stage. While authorization controls are necessary for governance and accountability, excessive dependence on individual decision-makers may delay service delivery when responsible officers are unavailable. Digital transformation principles recommend flexible workflow designs that maintain proper authorization while reducing unnecessary administrative delays (Mergel et al., 2019).

Furthermore, many existing digital permit studies focus mainly on application submission and document management while giving limited attention to field-level permit verification.

In municipal construction management, ensuring that approved permits correspond to actual construction sites is essential. Static paper permits or downloadable digital documents may still be vulnerable to duplication, alteration, or misuse. Integrating technologies such as QR

code verification provides opportunities for municipal inspectors to validate permit authenticity directly at construction locations.

The literature also reveals a gap regarding localized solutions for fragile and developing contexts. Advanced digital permitting systems implemented in developed countries often depend on sophisticated infrastructure such as BIM integration, high-speed internet connectivity, and complex enterprise platforms. Although these approaches provide significant benefits, they may not always be suitable for municipalities operating under resource constraints. (Heeks, 2003) emphasizes that digital systems in developing environments should be designed according to local technological conditions, organizational capacity, and user requirements.

In the context of Mogadishu Municipality, the field investigation conducted for this research identified several operational challenges that reflect issues discussed in the literature. The current process still requires applicants to visit municipal offices for application submission or staff assistance. Permit fee payments are recorded through manual evidence, and application approval depends on availability of authorized personnel. These limitations demonstrate the need for a more citizen-centered, flexible, and integrated digital solution.

Based on these identified challenges, the proposed Web and Mobile Digital Building Permit System aims to overcome existing limitations by introducing a synchronized digital platform that combines mobile citizen services with web-based municipal management. The system enables mobile application submission, digital payment recording during application submission, electronic receipts, single-level approval through authorized users, real-time application tracking, automated notifications, QR code permit verification, and centralized

permit data management. These improvements address both technological and operational gaps identified in previous research.

2.5 Research Gap

The literature reviewed in this chapter demonstrates that digital transformation has significantly improved public service delivery through the adoption of electronic government, mobile government, and digital information systems. Previous studies have shown that digital building permit systems enhance administrative efficiency by reducing paperwork, improving document management, supporting electronic workflows, and increasing transparency in municipal service delivery (OECD, 2020; United Nations, 2022). These developments have enabled many municipalities to modernize permit administration while improving communication between government institutions and citizens.

Despite these advancements, the literature also indicates that many municipalities, particularly in developing countries, continue to experience challenges during the implementation and operation of digital permit management systems. Several studies report that digital transformation often focuses primarily on internal administrative functions while many service delivery processes continue to rely on manual procedures, physical document handling, and face-to-face interactions between citizens and government officials (Heeks, 2002, 2003)

Furthermore, previous research has concentrated largely on advanced digital permitting technologies such as Building Information Modeling (BIM), Geographic Information Systems (GIS), cloud computing, and integrated digital government platforms (Eastman et al., 2011; Noardo et al., 2020).

Although these technologies provide significant benefits, they generally require sophisticated technological infrastructure, substantial financial investment, and high levels of organizational readiness. Consequently, many developing municipalities continue to operate partially digital permit systems that do not fully support citizen-centered digital service delivery. The field investigation conducted for this research revealed that the current building permit management process in Mogadishu Municipality has already undergone partial digital transformation. Compared with the previous paper-based approach, permit information is now managed electronically, improving record keeping and administrative organization. However, the investigation identified several operational challenges that continue to affect service efficiency and accessibility.

First, citizens generally cannot independently submit new building permit applications through digital platforms. Instead, applicants normally visit municipal offices where staff members enter application information into the existing system on their behalf. This limits the convenience and accessibility expected from modern digital government services.

Second, permit fee payment is performed during application submission, but applicants receive manual paper receipts as evidence of payment. Although payment collection is incorporated into the existing workflow, reliance on paper-based payment evidence limits digital financial record management and may reduce administrative efficiency.

Third, the current approval process depends on a single authorized approval before permit issuance. When the authorized officer responsible for approval is unavailable, application processing may be delayed, affecting the overall efficiency of municipal service delivery.

Finally, applicants have limited access to real-time information regarding the progress of their applications. Communication regarding application status often depends on direct

interaction with municipal staff rather than automated digital notifications or self-service tracking mechanisms.

These findings indicate that, although the existing system represents an improvement over traditional paper-based procedures, opportunities remain to further strengthen digital service delivery by improving citizen accessibility, simplifying administrative workflows, and enhancing transparency.

To address these identified gaps, this study proposes the design and development of a Web and Mobile Digital Building Permit System for Mogadishu Municipality. Unlike the existing process, the proposed system enables citizens to submit building permit applications directly through a mobile application while maintaining office-assisted services for applicants who prefer or require assistance from municipal staff. The system records permit payments digitally during application submission, generates electronic payment receipts, supports approval by either an authorized administrator or authorized staff member, provides real-time application status tracking, delivers automated notifications, and introduces QR code-based permit verification for improved authenticity and field inspection.

Therefore, this research contributes to the existing body of knowledge by proposing a practical and citizen-centered digital building permit solution that addresses operational challenges identified through both the literature review and field investigation. Rather than focusing on complex technological frameworks requiring advanced infrastructure, the proposed system emphasizes usability, accessibility, administrative efficiency, and compatibility with the operational environment of Mogadishu Municipality.

CHAPTER III: RESEARCH METHODOLOGY

3.1 Introduction

This chapter outlines the detailed system description, key features, architecture, development approach, hardware and software specifications, and user requirements for the proposed Building Permit System. Serving as the primary technical foundation for the study, this framework formalizes how the functional requirements of Mogadishu Municipality materialize into a multi-tier digital software solution. It connects the problem statement and research objectives identified in earlier chapters to the actual technical design and system implementation.

The methodology adopts the Agile Software Development Life Cycle (Agile SDLC) to manage the iteratively planned design, prototyping, testing, and implementation phases of the system. Agile was selected due to its flexibility, iterative nature, and focus on continuous feedback, which ensures that each functional module from document upload validation to automated permit verification meets institutional governance standards.

The structural blueprint of the system encompasses a multi-tier client-server architecture consisting of a React.js Web Administration Console (built with Vite), a Flutter-based Mobile Application, a Node.js and Express.js RESTful API runtime, and a MongoDB document-oriented database. The subsequent sections of this chapter provide a comprehensive breakdown of the structural components, data flow representations, unified modeling language (UML) diagrams, and database schemas that guarantee system integrity, role-based access control, and transaction auditing.

3.2 System Overview

The Building Permit System is a unified, cross-platform enterprise digital platform engineered to modernize and streamline the building permit authorization workflows for the Banadir Regional Administration. The system consolidates property registration, administrative document auditing, fee appraisal, and field permit verification into a centralized architecture to enhance operational transparency, prevent revenue loss, and eliminate permit forgery across Mogadishu Municipality.

The system operates across two interconnected software interfaces communicating through secure RESTful APIs:

- **Web Administration Console:** Developed using React.js (built with Vite) for Municipal Super Administrators and District Desk Officers. It serves as a centralized management portal providing administrative dashboards for application auditing, structural engineering review, fee validation, geographic permit approvals, audit logging, and system-wide reporting.
- **Mobile Application:** Developed using the Flutter framework for Citizens and Field Inspection Officers. It provides an intuitive interface enabling citizens to submit permit applications, upload land deeds, complete fee payments, and track approval statuses. Simultaneously, it empowers field officers to perform real-time, camera-based QR code verification of building permits on construction sites.
- **Backend Runtime:** Built with Node.js and Express.js, the backend manages application workflows, handles stateless JSON Web Token (JWT) authentication, enforces role-based access control (RBAC), and exposes secure API endpoints for web and mobile clients.

- Database Management System: Utilizes MongoDB, a document-oriented NoSQL database, to store structured property records, applicant metadata, payment transaction logs, user credentials, approval history, and dynamic QR verification tokens.

3.3 System Features

- User Authentication and Secure Profiles: Implements Role-Based Access Control (RBAC) to distinguish between Applicants, Municipal Administrators, and Field Inspectors while securing their profiles.
- Digital Document Submission Portal: Requires applicants to upload digital copies of their National ID/Passport and Title Deeds, completely eliminating the need for physical office visits.
- Integrated Revenue Processing Gateway: Automates the collection of permit fees by linking a mobile money transaction flow within the application submission line to ensure financial transparency.
- Real-Time Status Tracking: Allows applicants to view the current stage of their application, such as pending review, payment required, or approved, directly via their mobile interface.
- Context-Aware Resubmission Loop: Allows applicants to resubmit returned or rejected applications with corrections free of charge, unless building parameters (floors/area) are scaled up, which automatically calculates the balance fee.
- QR-Code Generation Engine: Generates a unique, encrypted QR code for every approved permit, acting as a digital signature that links the physical construction site to the database.

- Field Inspection & Verification: Equips inspectors with a mobile-native scanner interface to verify construction legality on-site with instant database feedback or manual ID search fallback.

3.4 System Architecture

The application leverages a classic Three-Tier Architecture model to isolate core operational concerns, maximize system scalability, and ensure secure data flow across all municipal nodes. This architectural structural framework separates user interaction, business rules, and persistent storage mechanisms into distinct intellectual layers.

- Presentation Layer (Client Tier): This layer handles user interaction and interface rendering across two specific operational channels. Citizens and field inspectors utilize a cross-platform Flutter mobile application to submit application metrics, track processing states, and scan security QR codes. Concurrently, high-level municipal officials and district staff access a responsive web interface built using the React framework to monitor district performance dashboards, approve permit structures, and audit overall administrative operations.
- Business Logic Layer (Application Tier): Serving as the core operational brain, this layer is built on top of a Node.js runtime environment leveraging the Express.js framework to orchestrate system functionalities. It exposes a unified RESTful API surface that processes inbound HTTP operations from both frontend instances. This tier handles authentication validations using cryptographic JSON Web Tokens (JWT) to secure endpoint boundaries, enforces role-based authorization parameters, parses multi-part form requests using Multer middleware layers, and maps automated operational updates based on municipal feedback.

- **Data Storage Layer (Data Tier):** This layer implements a split-persistence storage strategy designed to handle both structured operational metadata and physical electronic document uploads efficiently. Core textual data models, applicant profile schemas, tracking historical logs, and authentication parameters are managed inside a document-oriented MongoDB database. Conversely, actual physical file binaries, such as scanned National Passenger IDs, passports, and land title deeds, are pushed directly onto the host server's local file repository (/backend/uploads/) using programmatic unique string timestamps, storing only the reference string paths inside the corresponding MongoDB document objects to optimize overall storage querying speeds.

3.5 System Requirements Specification

3.5.1 Hardware Requirements

- **Server-Side**
 - Processor: Intel Core i5 or higher
 - RAM: Minimum 8 GB
 - Storage: minimum 256 GB SSD (scalable with demand)
- **Client Devices**
 - Smartphones: Latest Android and iOS Versions with GPS and internet support
 - Admin PCs: Core i5+, 8 GB RAM, SSD for smooth operation

3.5.2 Software Requirements

- Server Software
 - Operating System: Windows or Ubuntu
 - Database Engine: MongoDB
 - Web Server: Express.js Static Server
 - Runtime Environment: Node.js (v18 or higher)
- Client Software
 - Mobile: Flutter SDK, Android SDK, Dart
 - Admin Panel: Compatible browsers (Chrome, Firefox, Edge)
- Security
 - File Upload Security: Multer middleware for validation
 - Role-Based Access Control (RBAC): for sensitive operations
 - Authentication: JSON Web Tokens (JWT) & Bcrypt hashing
- Scalability & Maintenance
 - Codebase maintenance: via GitHub Repository
 - Database Backups: Cloud-based automated backups

3.6 Requirements Specification

3.6.1 Functional Requirements

- **User Authentication & Registration:** The system must allow users to register and login securely based on their roles (Applicants, Administrators, and Field Inspectors).
- **Digital Document Submission:** The system must allow applicants to upload scanned copies of their National ID/Passport and Land Title Deeds via the mobile application.
- **Live Application Tracking:** The system must allow applicants to view the real-time status of their permit application (Pending, Paid, Approved, or Rejected).
- **Application Resubmission Handling:** The system must allow applicants to update and resubmit returned applications based on staff rejection remarks without requiring duplicate base payments.
- **Integrated Payment Processing:** The system must support automatic mobile money billing to collect permit fees before pushing the file to the review queue.
- **Geographic District Filtering:** The system must allow municipal desk officers to view, filter, and audit application records specifically mapped to their respective districts in Mogadishu.
- **QR-Code Generation & Scanning:** The system must automatically generate an encrypted QR code for every approved permit and allow field inspectors to scan and verify it using their mobile devices.

3.6.2 Non-Functional Requirements

- **Security & Encryption:** All user passwords must be securely hashed using Bcrypt before being stored in MongoDB, and administrative routes must be protected via JSON Web Tokens (JWT).
- **Performance & Response Time:** The system backend should process API request operations within less than 2 seconds under normal network conditions.
- **Usability & Accessibility:** The mobile application interface must be simple, easy to navigate, and include a light/dark mode switch to prevent eye strain.
- **Reliability & Backups:** The system data must remain highly available, leveraging cloud-based automated database backups to prevent data loss.
- **Maintainability:** The codebase must be modularly structured and hosted on GitHub to ensure smooth updates and easier bug tracking.

3.7 Feasibility Study

3.7.1 Technical Feasibility

The project is highly feasible technically as it utilizes free, open-source, and widely supported modern technologies. The MERN stack (MongoDB, Express.js, React, Node.js) and Flutter SDK have extensive documentation, making development, integration, and debugging highly manageable within the current infrastructure.

3.7.2 Economic Feasibility

The development costs are virtually zero since all the development tools, frameworks, and database packages used are open-source and free. The system

does not require expensive high-end hardware, and future hosting costs on cloud services like MongoDB Atlas can be scaled progressively based on municipal budget availability.

3.7.3 Operational Feasibility

The system is highly operational and user-friendly for both citizens and municipal staff. The mobile application features a simple interface requiring minimal training for applicants and inspectors, while the centralized web dashboard fits smoothly into the existing district review structures of the Banadir Regional Administration.

3.7.4 Schedule Feasibility

The project schedule is fully realistic and achievable, structured across a 4-month development timeline divided into weekly milestones. The timeline allocates Month 1 for requirements gathering and system architecture design, Month 2 for database modeling and backend API development, Month 3 for cross-platform mobile and web frontend engineering, and Month 4 for system integration, rigorous testing, and final thesis deployment before the official defense date.

3.8 UML Diagrams

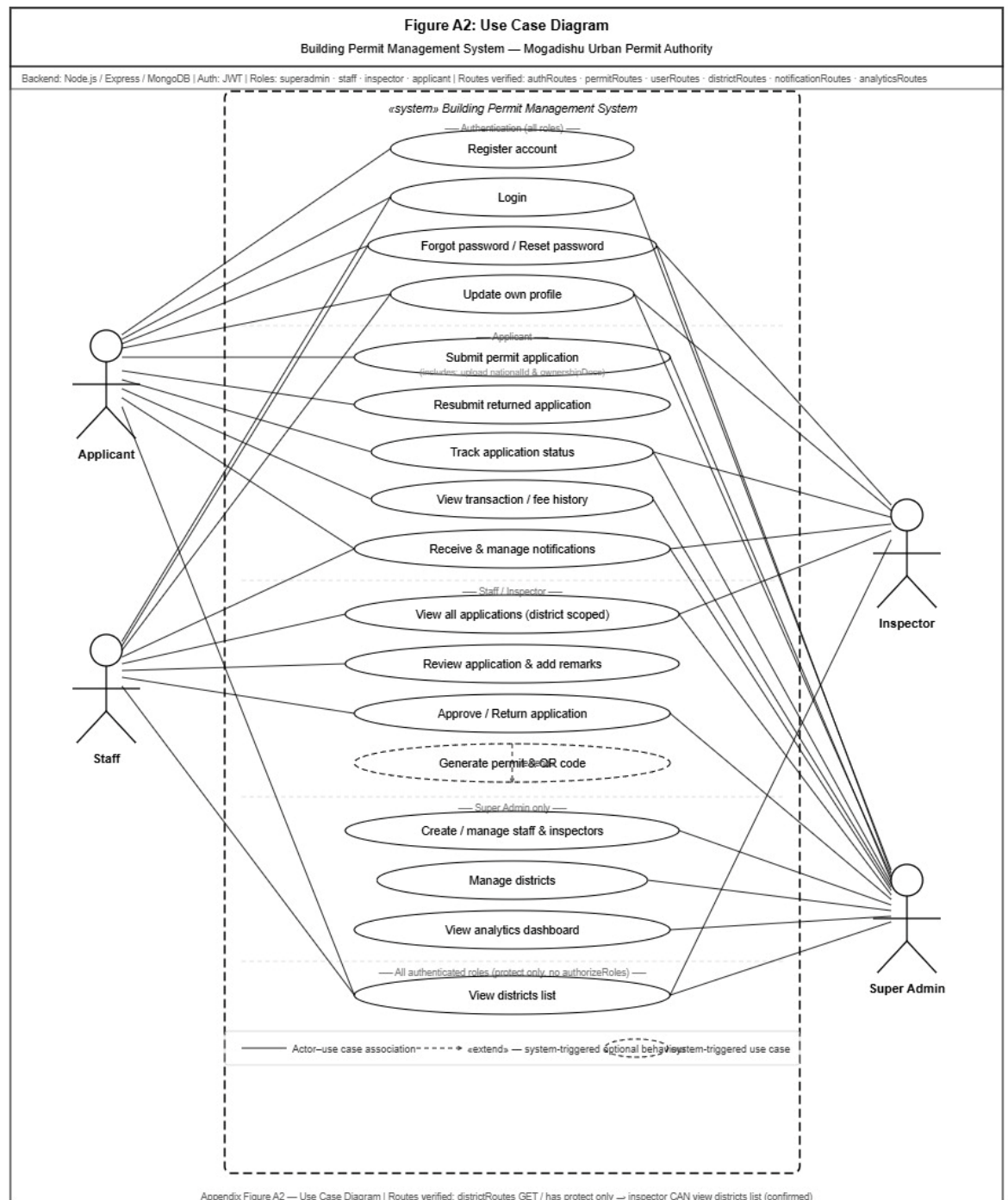


Figure 3.1: UML Diagrams

3.9 Database Design

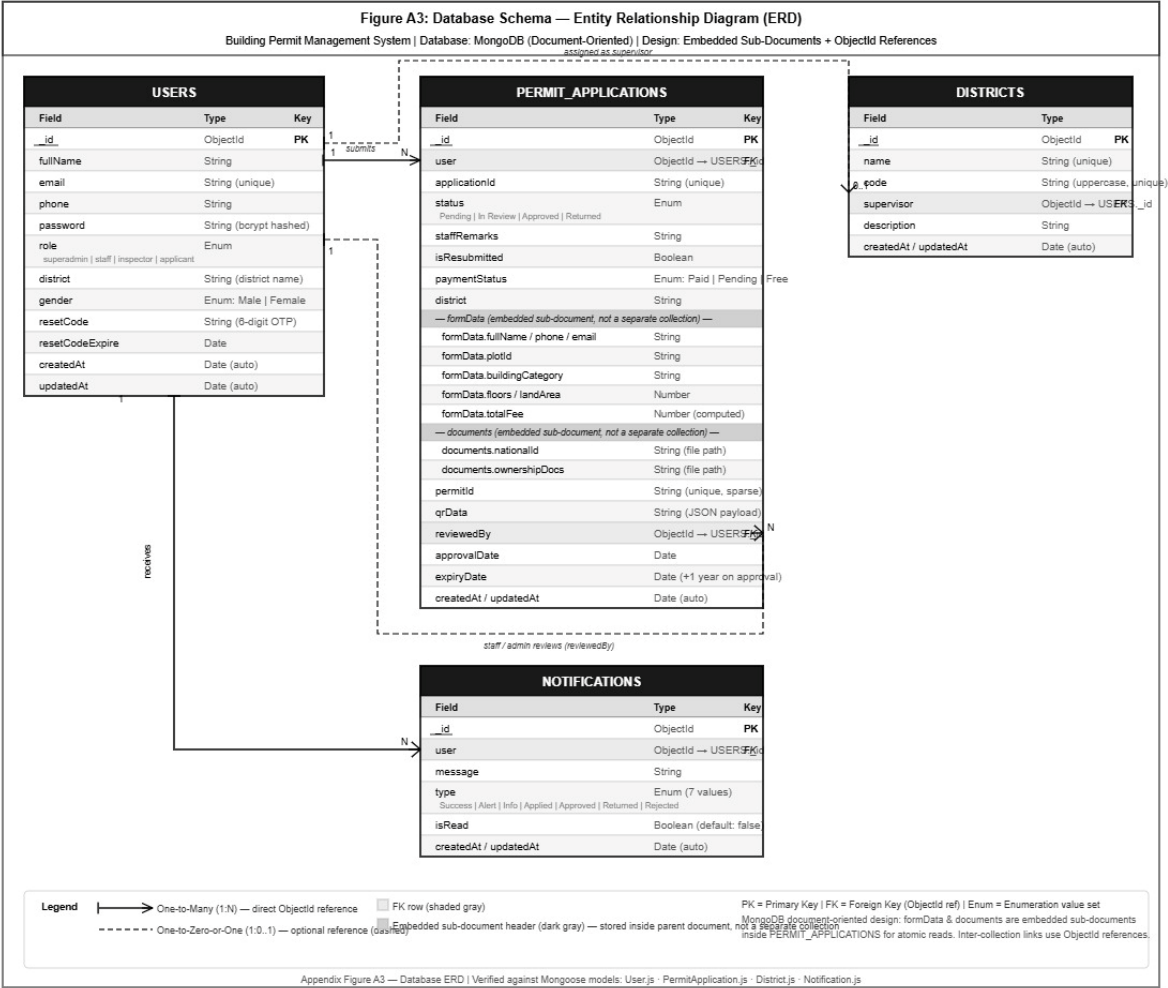


Figure 3.2: Database Design

CHAPTER IV: IMPLEMENTATION AND TESTING

4.1 Introduction

This chapter presents the implementation details and experimental evaluation results of the Building Permit System. It articulates how the architectural designs, entity relationships, and system workflows formulated in the preceding chapter were realized into a production-ready software solution. The core software ecosystem integrates four primary technology pillars: a responsive React.js Web Administration Portal, a cross-platform Flutter Mobile Application, a scalable Node.js/Express.js Backend API, and a document-oriented MongoDB Database. Together, these components establish a unified digital infrastructure for managing municipal permit submissions, document verifications, field inspections, fee processing, and administrative auditing.

Furthermore, this chapter documents the formal verification procedures executed to confirm that the developed platform fulfills both functional specifications and operational constraints. Comprehensive testing frameworks encompassing functional black-box evaluation, user interface (UI/UX) assessment, mobile integration testing, device compatibility checks, and security vulnerability validation were performed. High-resolution interface screenshots are embedded throughout the text to illustrate actual system performance and user interaction workflows under operational conditions.

Ultimately, the findings presented in this chapter demonstrate that the Building Permit System effectively automates paper-based municipal procedures, mitigates administrative delay, enhances transparency, and delivers a robust digital platform tailored for urban governance.

4.2 Web Admin Panel Implementation and Testing

The Web Admin Panel serves as the centralized command and control center for the Building Permit System. It equips Super Administrators, District Managers, and Municipal Review Officers with the tools required to oversee permit workflows, configure municipal zones, review uploaded legal deeds, assign field inspectors, and analyze fee collection analytics.

The admin console was constructed using React.js for modern state management, styled with custom administrative UI components, and integrated with a Node.js/Express.js API server connected to MongoDB. The layout follows an intuitive dashboard architecture designed to maximize information density while minimizing cognitive load for operational personnel.

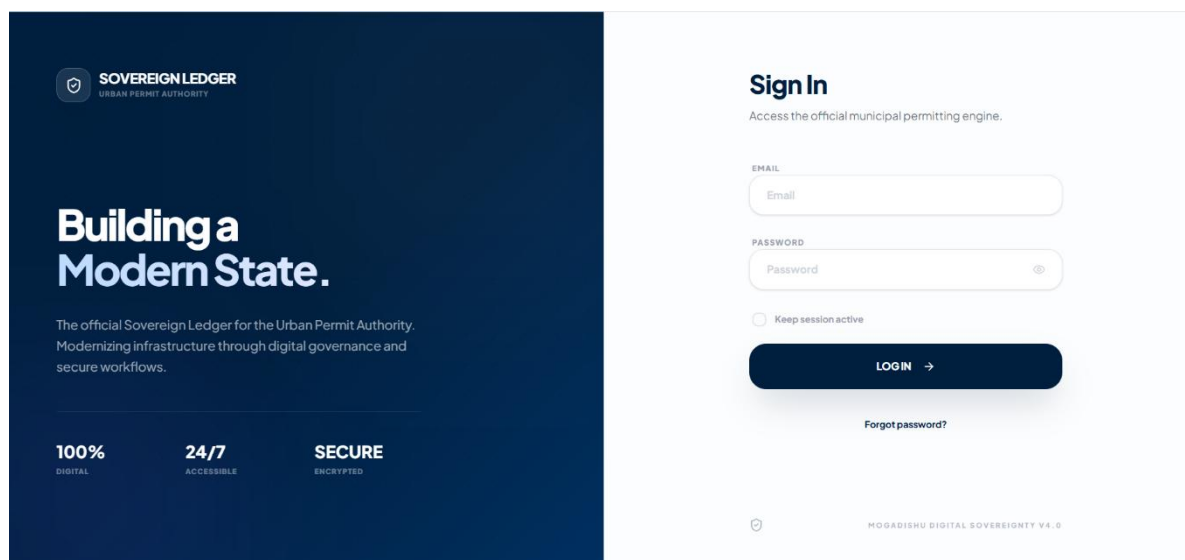


Figure 4.1: Administrative Login Screen.

The login page provides secure authentication for authorized users before granting access to the administrative control panel. The interface is designed with an institutional layout that reinforces system identity while maintaining strict security standards.

The login module authenticates administrators and municipal staff using encrypted credentials before allowing entry into the platform. Users enter their registered email address

and password, with options for password visibility toggling and session persistence (Keep session active). After successful authentication, the backend validates user roles and automatically redirects users to their respective administrative dashboards.

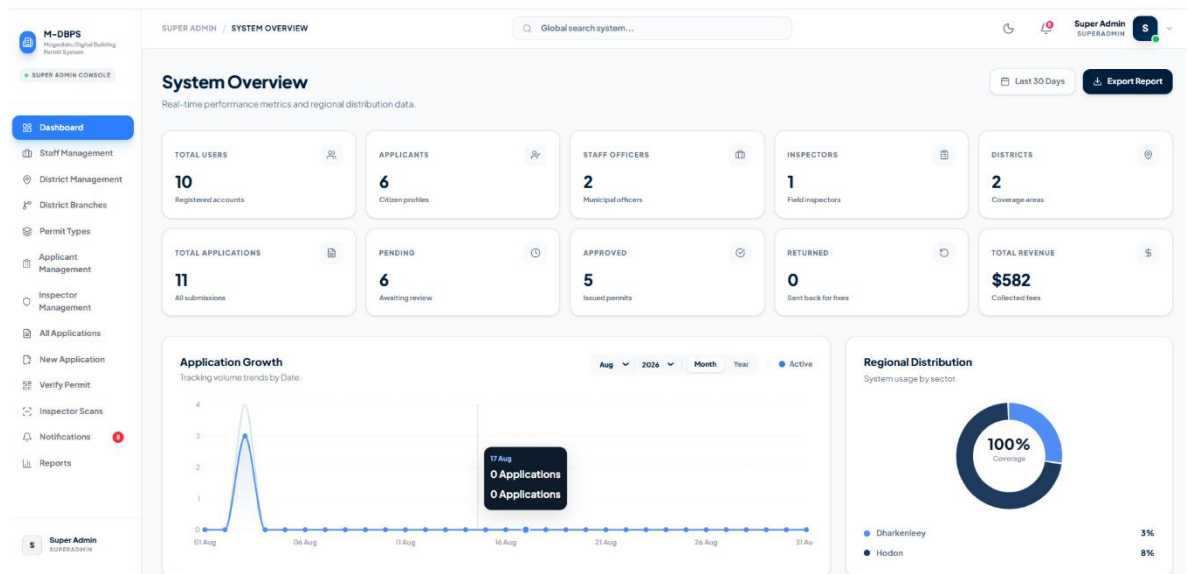


Figure 4.2: Super-Admin System Overview & Administrative Control Dashboard.

The Super-Admin Dashboard provides a comprehensive overview of critical system metrics, financial data, and regional usage statistics. It displays key performance indicators including total registered users, overall permit revenue collected, pending applications requiring immediate review, and total approved building permits. This enables executive administrators to monitor municipal operations from a single centralized workspace.

In addition to summary metrics, the dashboard features interactive analytics such as the Application Growth trend line and the Regional Distribution sector chart. A persistent navigation sidebar on the left side of the screen provides direct access to all core system

modules, including Staff Management, District Branches, Building Categories, Applicant Verification, Inspector Assignment, Application Audits, and Reports.

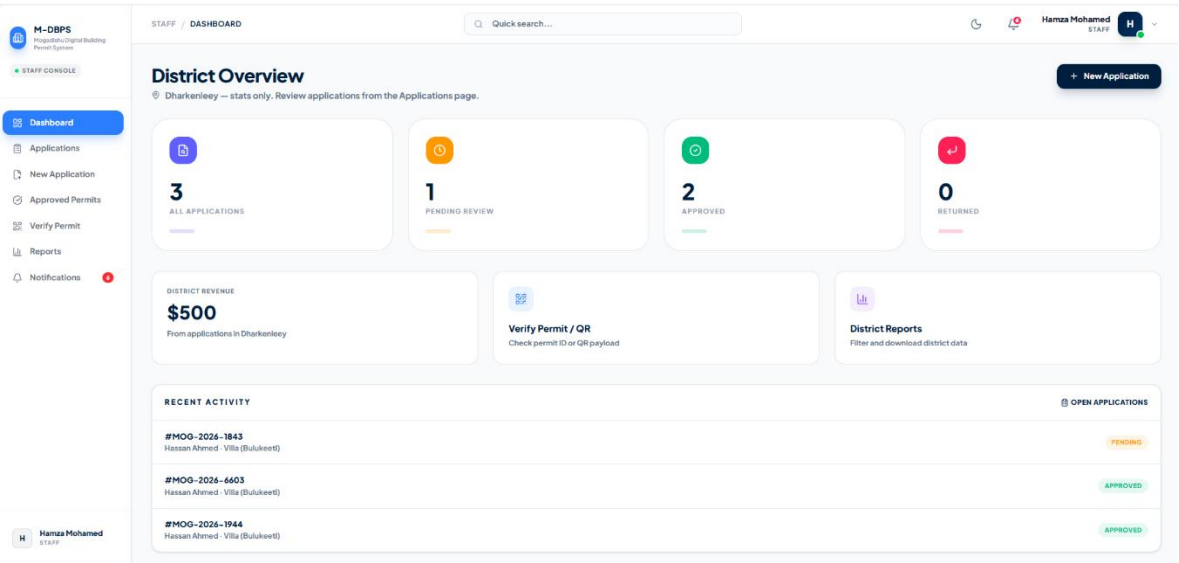


Figure 4.3: Municipal Staff Operational Dashboard & Verification Portal.

The Staff Dashboard serves as the primary workspace for municipal review officers responsible for processing permit applications. It summarizes daily workloads using dynamic metric cards that categorize applications into Total Applications, Total Pending, Approved, and Returned statuses. This structure allows staff members to prioritize urgent review tasks efficiently.

Below the summary metrics, the dashboard contains an interactive Applications Data Table that displays detailed applicant records, including unique Application IDs, applicant names, contact phone numbers, requested permit types, and submission dates. Staff officers can utilize action triggers on each row, such as the Perform Review button or the quick-view inspection icon, to evaluate uploaded property deeds and make informed approval or return decisions.

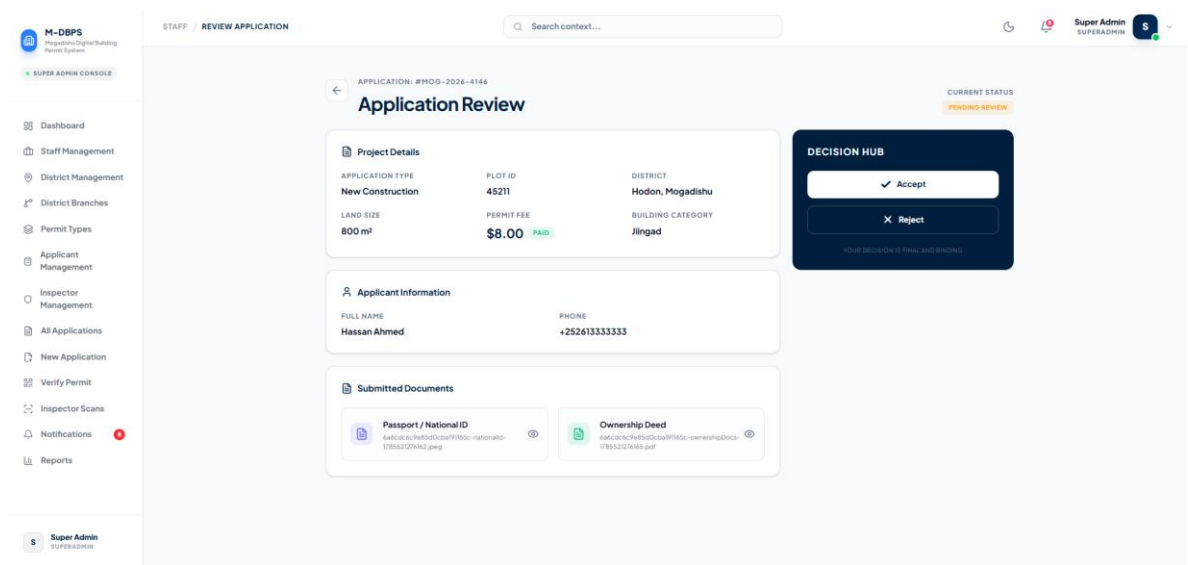


Figure 4.4: Application Review and Decision Interface

The Application Review interface provides a centralized workspace for municipal staff to evaluate submitted permit requests. It consolidates key project parameters, applicant metadata, and uploaded legal documents into a single screen, streamlining the review workflow without requiring multi-tab navigation.

The interface displays project parameters such as Plot ID, district location, land size, building category, and fee payment status alongside verified applicant contact details. Reviewing officers can preview uploaded identification and land deeds directly within the browser before using the Decision Center panel to formally Accept or Reject the permit application.

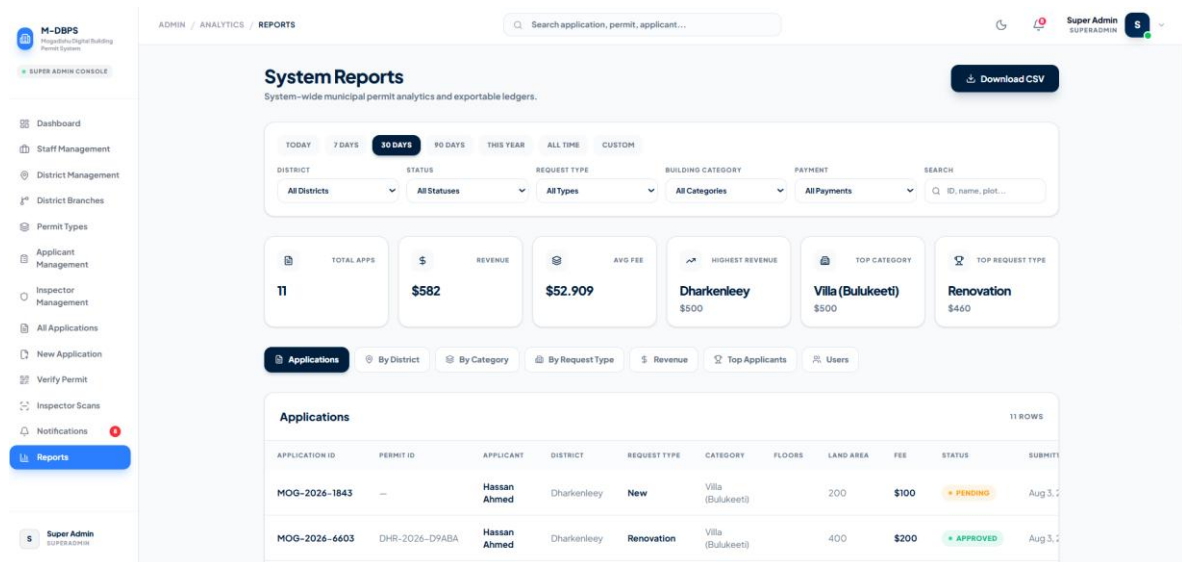


Figure 4.5: Reports and Analytics Dashboard

The Reports and Analytics module provides administrators with a comprehensive statistical ledger of municipal operations and fee collections. The dashboard displays high-level operational metrics including registered users, actual active applicants, and average processing speed alongside visual charts for monthly revenue trends and district application distributions. Additionally, a detailed breakdown table categorizes application volume, processing efficiency, conversion rates, and financial yield across municipal sectors, complete with export options (Export to PDF/Excel) for administrative auditing.

4.2.1.1 Functional Testing

Functional testing was conducted to verify that every major function of the Building Permit System operates according to the specified system requirements. The testing focused on validating user interactions, administrative control flows, data processing, document verifications, fee calculation logic, and role-based permissions between Super Administrators and Municipal Staff Officers. A black-box testing approach was adopted,

where the functionality of each module was tested by providing different inputs and observing the outputs without considering the internal source code.

The testing strictly covered the major components of the Web Administration Console, including user authentication, staff provisioning, district and branch configuration, structural building type setup, applicant verification, application review workflows (Accept/Reject decision processing), inspector assignments, and dynamic financial/operational reporting. Each function was tested using valid and invalid input data to ensure that the system handled different operational scenarios correctly.

The test results demonstrated that the developed web system successfully performed all essential operations while providing accurate responses and maintaining data integrity across MongoDB and API endpoints.

Table 4.1: Functional Testing

Test ID	Module	Test Scenario	Expected Result	Status
TC-01	Authentication	Login with valid Super-Admin credentials	User is authenticated and redirected to the System Overview Dashboard	Passed
TC-02	Authentication	Login with invalid password	Error message is displayed and administrative access is denied	Passed

TC-03	Staff Management	Provision a new municipal staff account	Staff profile is created and assigned to a specific district console	Passed
TC-04	District Management	Register a new municipal district and branch	District record (e.g., Dharkenleey) is saved and linked to system workflows	Passed
TC-05	Building Types	Create building category with fee parameters	Category (e.g., Jiingad) is stored and configured for permit fee generation	Passed
TC-06	Applicant Management	Review citizen profile and contact details	Verified applicant credentials (+252...) and history are displayed accurately	Passed
TC-07	Document Review	Inspection of attached legal deeds and IDs	Inline document preview modal renders uploaded PDF/Image files	Passed
TC-08	Application Review	Formally 'Accept' an audited permit request	Application status updates to APPROVED and generates approval metrics	Passed
TC-09	Application Review	Formally 'Reject' an incomplete application	Application status changes to RETURNED/REJECTED requiring applicant correction	Passed

TC-10	Inspector Management	Assign a field inspector to property site	Inspection task is logged and mapped to the assigned field officer	Passed
TC-11	Applications Register	Filter applications by district or review state	Data table dynamically renders matching records (e.g., #MOG-2026-9734)	Passed
TC-12	Reports & Analytics	Filter monthly revenue and sector distribution	Visual charts and conversion rate tables refresh with active fiscal data	Passed
TC-13	Reports Export	Execute report export action	System compiles and exports financial/performance metrics to PDF/Excel	Passed
TC-14	Role-Based Access	Staff user attempts Super-Admin action	System blocks unauthorized operation based on RBAC rules	Passed
TC-15	Data Validation	Submit missing required configuration fields	Form submission is blocked with explicit visual validation alerts	Passed

The functional testing results indicate that all major system modules operated as expected. Each feature successfully completed its intended task without producing critical errors during testing. The role-based access mechanism correctly restricted users to their authorized functions, while the validation mechanisms prevented invalid data from being stored in the database.

Overall, the functional testing confirms that the Building Permit System satisfies its functional requirements and is capable of supporting municipal permit auditing, document verification, staff operations, decision processing, and executive reporting in a reliable and efficient manner.

4.2.1.2 User Interface (UI/UX) Evaluation

The user interface of the Building Permit System was engineered to provide a professional, transparent, and highly accessible experience for municipal personnel. Since the web platform is utilized by different operational roles including Super Administrators, Municipal Staff Review Officers, and District Managers special design consideration was directed toward layout consistency, information density, and ease of navigation throughout the development cycle.

The Web Administration Console adopts a clean dashboard framework structured with an intuitive persistent sidebar navigation menu, dynamic KPI metrics summary cards, responsive data tables, clear status badges (APPROVED, PENDING, RETURNED), and interactive verification modals. Reviewing officers can navigate between district branches, application rosters, and reporting suites without procedural complexity. Typography, modern icons, structural spacing, and institutional color palettes (navy blue, gold accents, and

context-specific status indicators) were carefully curated to maximize readability and reduce human review errors.

To elevate user experience, the web application delivers immediate feedback upon every administrative transaction. Contextual toast notifications, inline validation warnings, dynamic fee preview cards, and confirmation dialogs such as explicit prompts when executing an application "Accept" or "Reject" decision ensure that staff members clearly understand the operational status of their actions.

During the user testing phase, the interface was evaluated by municipal administrative personnel. Testers reported that the navigation was intuitive and required minimal system orientation. Their user feedback drove several targeted refinements, including simplified inline document previews for property deeds, clearer action button labels in the decision workspace, dynamic table filtering controls, and enhanced responsiveness across desktop and laptop display viewports.

Overall, the UI/UX evaluation confirmed that the developed interface fulfills usability requirements and facilitates seamless, error-free interaction between municipal staff and the Building Permit System.

4.2.1.3 Summary of Web Application Testing

Following the implementation of the Web Administration Console, rigorous end-to-end testing was conducted to verify that every functional module operated accurately across various operational constraints. The testing strictly evaluated administrative capabilities, encompassing user authentication, staff account allocation, district and branch configuration, structural building category setup, applicant verification, application auditing, legal

document inspection, decision workflow execution (Accept/Reject), financial analytics processing, and audit logging.

The empirical results indicated that all administrative functions performed successfully with high data integrity and robust validation controls. Role-Based Access Control (RBAC) mechanisms effectively restricted staff users to their designated regional consoles, preventing unauthorized entry to executive configuration utilities. Simultaneously, the MongoDB database and RESTful APIs processed, updated, and retrieved system records without data corruption or loss.

The technical responsiveness and layout adaptation of the web application were also evaluated across various screen resolutions and modern desktop browsers (Google Chrome, Mozilla Firefox, Microsoft Edge). The interface maintained stability, responsive element alignment, and visual performance throughout all testing procedures.

Based on the aggregated test outcomes, the Web Administration Console fulfilled all functional, operational, and interface specifications, proving reliable, secure, and fully prepared for operational deployment within the Mogadishu Municipality administrative ecosystem.

4.2.2 Mobile Application Implementation and Testing

The Mobile Application represents the primary client interface used by citizen applicants and municipal field inspectors to interact with the Building Permit System. It was developed using the Flutter framework, enabling the application to operate efficiently across mobile devices while providing a responsive, cross-platform, and consistent user experience.

The mobile application communicates seamlessly with the backend ecosystem through secure RESTful APIs developed using Node.js and Express.js. Core operational records and

application payloads have persisted in MongoDB, while integrated notification services deliver real-time application status updates directly to users' mobile devices.

The mobile application simplifies municipal permitting procedures by allowing citizens to register profile credentials, initiate new permit requests, calculate dynamic appraisal costs, upload property ownership deeds, track application processing states (PENDING, APPROVED, RETURNED), and display official digital permit QR codes directly from their smartphones. Concurrently, field inspectors utilize the application to scan permit QR codes on-site, verify property compliance, and update inspection records in real time. The implementation focused on interface simplicity, data encryption, and high performance to ensure that citizens and field officers can complete their tasks rapidly and reliably.

4.2.2.1 Mobile Application Features

The Building Permit System Mobile Application provides several vital features designed to streamline municipal service delivery and field verification operations. These features are structured according to the functional responsibilities assigned to citizen applicants and field inspection personnel.

The major features integrated into the mobile application include:

- Secure Authentication & Profile Setup.
- Interactive Permit Application Submission.
- Real-Time Appraisal Fee Calculation.
- Document Attachment & Upload Module.
- Live Application Status Tracking.
- Digital Permit & QR Code Generator.

- Field Inspection & QR Code Verification.
- Real-Time System Notifications.
- User Profile & Activity History Management.

Each feature was implemented to eliminate manual bureaucratic overhead, replace paper-heavy application channels, and increase transparency between Mogadishu Municipality and citizen property developer.

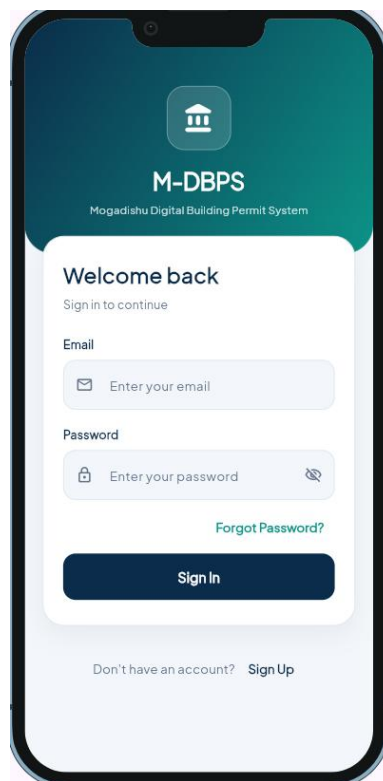


Figure 4.6: Mobile Authentication Interface.

The Mobile Authentication interface enables citizen applicants and field inspectors to securely access the Building Permit System using their registered credentials. The login interface incorporates username or email-based authentication paired with password verification. Upon successful credential validation, the authentication module secures API

communications via JSON Web Tokens (JWT) and redirects users to their designated mobile dashboard according to their assigned system role.

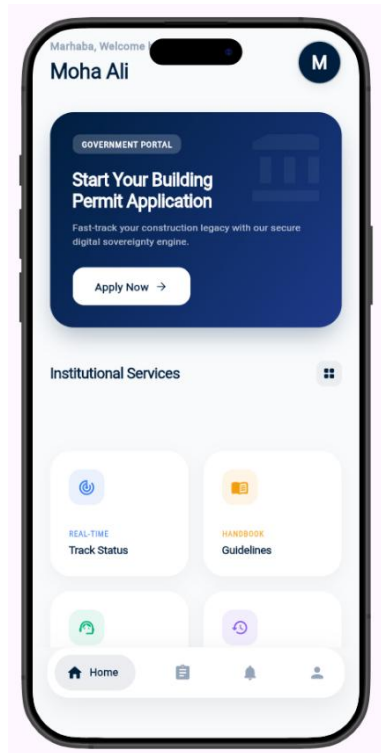


Figure 4.7: Mobile Applicant Dashboard

The Mobile Applicant Dashboard serves as the primary navigation hub after successful login. It presents citizens with quick access to municipal permitting services, active application summaries, and real-time status indicators. From this single screen, applicants can initiate new permit requests, calculate appraisal fees, monitor submitted applications through dynamic status badges (PENDING, APPROVED, RETURNED), and view generated digital permit QR codes. The interface is optimized for single-thumb navigation to ensure high accessibility and ease of use.

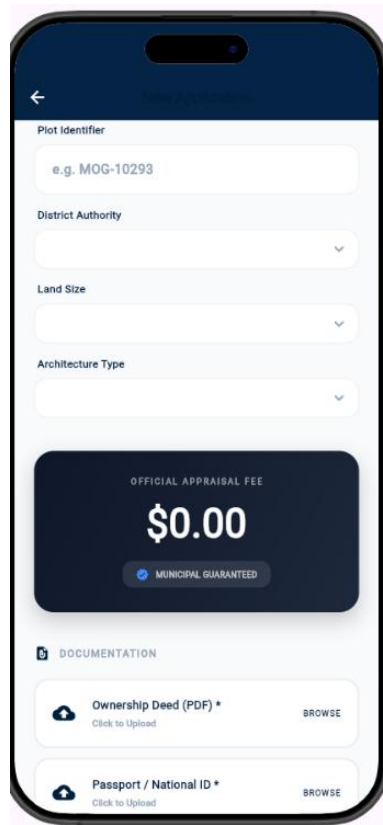


Figure 4.9: Permit Application Submission and Document Upload Screen

The Permit Application Submission interface enables citizen developers to register new construction projects by entering plot dimensions, selecting district locations, and defining structural building categories. The screen integrates an automated appraisal fee calculator that dynamically estimates permit costs prior to submission. Additionally, an integrated file picker allows applicants to attach essential legal documentation, including identity cards and land ownership deeds, ensuring that all necessary verification files are transmitted securely to municipal staff.

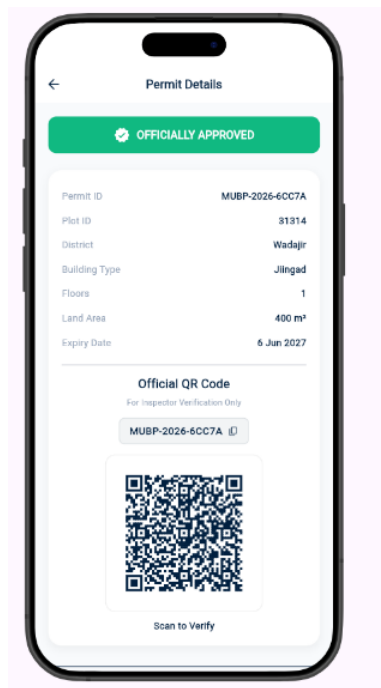


Figure 4.10: Digital Permit Details and Verification QR Code Screen

The Digital Permit Details interface serves as the official proof of municipal approval issued to citizen property owners upon successful review. The screen highlights an explicit status badge (OFFICIALLY APPROVED) alongside key structural and legal metadata, including the unique Permit ID (e.g., MUBP-2026-6CC7A), Plot ID, municipal district location, approved building classification, total floor count, land area measurements, and the permit expiration date.

Furthermore, the interface embeds an official dynamically generated QR Code labeled specifically for inspector verification. When municipal field inspectors scan this digital badge on-site using their mobile devices, the system cross-references the embedded permit identifier directly against MongoDB records to validate compliance and prevent construction fraud.

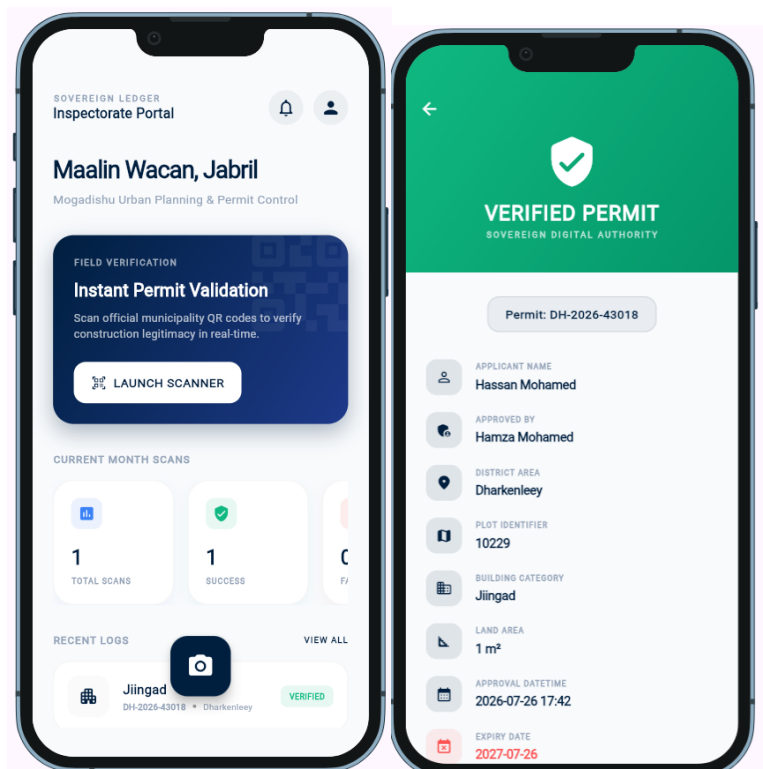


Figure 4.11: Field Inspector Portal and Permit Verification Workflow

The Field Inspectorate interface equips municipal officers with real-time verification tools to validate active construction sites directly from the field. The main Inspectorate Portal (left) features an operational dashboard displaying monthly scan metrics, historical audit logs, and a prominent trigger to launch the camera-based QR verification scanner. Upon scanning a site permit or manually querying a permit code, the system cross-references the spatial and legal data with MongoDB records to return the Verified Permit Screen (right). This result view displays a verified status header alongside key property records, including the applicant's name, approving municipal staff officer, district location, plot ID, structural building category, land measurements, approval timestamps, and expiration parameters to ensure complete construction legitimacy on-site.

4.2.2.2 Unit and Integration Testing

After completing the development of the mobile application, unit testing and integration testing were performed to verify the correctness of individual Flutter components and the seamless interaction between different client-server modules. Unit testing focused on evaluating each user interface module independently, while integration testing verified that data flow and end-to-end communication between the Flutter Mobile Application, Express.js REST API backend server, and MongoDB database functioned accurately under real-world network conditions.

Testing covered core operational workflows, including user authentication (email/username and password validation), permit application form submission, dynamic appraisal fee estimation, property deed attachment, QR code camera scanning, field verification queries, and real-time application status updates.

The results indicated that all mobile modules communicated successfully with the Express.js API endpoints, and permit payloads were accurately persisted and retrieved from the MongoDB database. Error handling, network timeout resilience, and input validation mechanisms also performed correctly when invalid parameters or malformed file uploads were introduced.

Table 4.2: Unit and Integration Testing

Test ID	Module	Test Scenario	Expected Result	Status
UT-01	Authentication	Login with valid user credentials	User is authenticated via JWT and redirected to dashboard	Passed

UT-02	Authentication	Submit login form with empty input fields	Visual validation alert is displayed and request is blocked	Passed
UT-03	Fee Calculator	Enter plot dimensions and building category	Dynamic appraisal fee (\$) is calculated and displayed accurately	Passed
UT-04	File Picker	Attach identity document and land deed	File payload (PDF/PNG) is validated for size and type	Passed
UT-05	QR Scanner	Launch camera verification component	Device camera opens and initializes scanner viewfinder overlay	Passed
IT-01	Permit Submission	Submit complete permit application form	Application payload is posted to Express API and saved in MongoDB	Passed
IT-02	Document Upload	Upload property deed during application creation	Document is securely stored and linked to applicant record ID	Passed
IT-03	Field Verification	Scan valid permit QR code on-site	Mobile app queries backend API and displays verified permit details	Passed

IT-04	Status Synchronization	Fetch live application progress state	Applicant dashboard updates status badge (PENDING/APPROVED)	Passed
IT-05	Alert Notification	Administrative decision updated on web admin	System delivers real-time status alert notification to mobile app	Passed

The testing results confirmed that the integration between the Flutter Mobile Application, Node.js backend runtime, Express.js API layer, and MongoDB database operated successfully. The mobile application demonstrated reliable performance, robust client-server synchronization, and fulfilled all technical requirements specified during the system engineering phase.

4.2.2.3 Device Compatibility Testing

Device compatibility testing was executed to evaluate the adaptive interface responsiveness, camera hardware API integration, and operational stability of the Flutter mobile client across diverse Android hardware configurations. Because citizen applicants and municipal field inspectors utilize smartphones with varying screen dimensions, display pixel densities, processing capacities, and Android OS releases, comprehensive compatibility verification was vital to ensure uniform service availability across all target user environments.

The evaluation specifically targeted dynamic layout reflows, hardware camera performance during QR code scanning, form input accessibility, real-time API request execution, and visual status badge rendering. Each physical device and virtual runtime environment was audited to verify structural alignment, smooth touch response rates, and robust backend synchronization.

The empirical results confirmed that the mobile client achieved flawless cross-platform rendering and layout scaling across all test hardware setups without layout clipping or operational degradation. Flutter's underlying layout engine automatically adjusted screen viewports to maintain optimal font legibility and functional usability across various physical aspect ratios.

Table 4.3: Device Compatibility Testing

Device Model	Operating System	Display Size	Evaluation Outcome
Samsung Galaxy A15	Android 14	6.5 inches	Passed
Tecno Spark 20	Android 13	6.6 inches	Passed
Infinix Smart 8	Android 13	6.6 inches	Passed
Xiaomi Redmi Note 13	Android 14	6.67 inches	Passed
Android Virtual Device (AVD)	Android 14	Emulator Viewport	Passed

The compatibility evaluation confirmed that the mobile application delivers consistent system functionality across diverse hardware environments. Core operations including user authentication, real-time permit appraisal calculations, camera-based QR code verification, legal document uploads, and REST API synchronization performed reliably throughout all hardware verification cycles.

4.3 Security Implementation and Testing

Security represents a foundational pillar of the Building Permit System due to the critical nature of stored municipal data, including citizen credentials, property ownership deeds, national identification records, land plot coordinates, structural appraisal transactions, and official approval logs. To guarantee data confidentiality, system integrity, and service availability across both web and mobile environments, multi-layered security protocols were engineered throughout the system architecture.

The security infrastructure integrates encrypted user authentication, granular Role-Based Access Control (RBAC), cryptographically salted password hashing, multi-tier input sanitization, stateless API communication security, and strict document upload validation. Security testing was subsequently executed to evaluate the efficacy of these protection mechanisms under various threat vectors and unauthorized access attempts.

4.3.1 Authentication and Authorization

Authentication enforces identity verification before granting access to protected municipal modules. Every user whether accessing the React.js Web Admin Console or the Flutter Mobile Application must present valid credentials. Upon successful credential validation, the Node.js/Express.js backend generates a cryptographically signed JSON Web Token (JWT). This token is transmitted in the HTTP Authorization header (Bearer <token>) to stateless rely subsequent API calls between client interfaces and server endpoints.

System authorization is governed by Role-Based Access Control (RBAC). Each user profile is bound to an explicit role with restricted operational privileges:

- Super Administrators: Possess full system privileges, including staff provisioning, district branch setup, structural building category management, and global financial report access.
- Municipal Staff Officers: Authorized to review permit applications, audit attached legal deeds, execute decision state changes (Accept or Reject), and manage regional application registers.
- Field Inspectors: Restricted to camera-based permit QR code scanning, property site verification, and field audit logging via the mobile interface.
- Citizen Applicants: Authorized to submit construction permit applications, calculate appraisal fees, upload ownership documents, track application progress, and display approved digital permit QR badges.

To protect user credentials against data breaches or database compromises, raw passwords are never persisted in plain text. Passwords are cryptographically hashed using the Bcrypt algorithm with an adaptive salt factor prior to storage within the MongoDB database.

4.3.2 Input Validation and Sanitization

Input validation mechanisms were implemented across both client and server layers to ensure data integrity and neutralize malicious input vectors.

Client-side validation is enforced within the React.js Web Console and Flutter Mobile App to deliver instantaneous user feedback, prevent empty form dispatches, and enforce expected field formats (such as telephone number structures, alphanumeric plot IDs, and numerical land area dimensions).

On the server side, the Express.js middleware executes rigorous request sanitization before database operations occur. API payloads are validated against strict structural schemas to protect against NoSQL query injections, cross-site scripting (XSS), and parameter tampering. Furthermore, file upload controllers enforce strict MIME-type checking and file size bounds on attached property deeds and identity documents, rejecting unauthorized executable files or unsupported extensions.

4.3.3 Security Testing

Security testing was conducted to evaluate the operational resilience of the implemented defense layers against common application vulnerabilities and privilege escalation attempts. Testing procedures evaluated authentication edge cases, unauthorized API calls without valid JWT headers, role-based boundary cross-overs, invalid parameter handling, and QR verification tampering. The primary objective was to confirm that restricted administrative assets remained impenetrable to unauthenticated actors while preserving data accuracy.

Table 4.4: Security Testing Results

Test ID	Security Scenario	Expected System Response	Status
ST-01	Login attempt with invalid password credentials	Access denied with explicit generic error payload	Passed
ST-02	Citizen role attempting access to Super-Admin endpoints	Access blocked with HTTP 403 Forbidden response	Passed
ST-03	Invoking protected REST API endpoints without JWT header	Request rejected with HTTP 401 Unauthorized state	Passed

ST-04	Submitting malformed JSON parameters to permit API	Payload rejected by middleware validation schema	Passed
ST-05	Scanning an expired or forged building permit QR code	System flags invalid payload and displays visual warning	Passed
ST-06	Uploading unsupported file format (.exe / .bat) as land deed	File controller blocks execution and rejects payload	Passed
ST-07	Intercepting and altering JWT payload parameters	Signature validation fails and token is invalidated	Passed
ST-08	Attempting NoSQL injection via search input parameters	Query sanitizer neutralizes operator symbols	Passed

The security evaluation verified that authentication, authorization, input sanitization, and access control mechanisms reliably guarded the application against unauthorized operations and data manipulation. No critical vulnerabilities were identified during the testing phase.

Overall, the Building Permit System satisfies all foundational security benchmarks expected of an enterprise-grade municipal governance platform.

4.4 Chapter Summary

This chapter presented the comprehensive engineering, architectural deployment, and multi-tiered testing evaluation of the Building Permit System for Mogadishu Municipality. The full-stack implementation integrated a React.js Web Administration Console, a Flutter Cross-Platform Mobile Application, and a Node.js/Express.js RESTful API runtime linked to a MongoDB database. Key functional modules were successfully operationalized, including administrative staff provisioning, district branch management, structural building category

setup, citizen profile registration, dynamic appraisal fee estimation, legal document verification, camera-based QR code field auditing, formal review decision workflows (Accept / Reject), and real-time revenue analytics.

Rigorous empirical testing was conducted across all system tiers to evaluate functional correctness, UI/UX usability, mobile device compatibility, and system security resilience. The evaluation confirmed that all web and mobile components performed seamlessly, enforcing Role-Based Access Control (RBAC), data sanitization, and stateless JWT authorization without critical errors. Ultimately, the Building Permit System delivers a secure, transparent, and efficient digital infrastructure that modernizes municipal governance, eliminates manual paper bureaucracy, and ensures construction legitimacy across Mogadishu Municipality, establishing a solid foundation for the final conclusions in the subsequent chapter.

CHAPTER V: DISCUSSION AND CONCLUSION

5.1 Introduction

This final chapter synthesizes the outcomes, engineering accomplishments, and operational insights derived from the design, development, and testing of the Digital Building Permit System for Mogadishu Municipality under the Banadir Regional Administration. It interprets the project's empirical results against its original goals, evaluates the achievement of the research objectives, details structural limitations, and outlines practical recommendations for future architectural iterations within municipal governance.

5.2 Discussion

The primary objective of this project was to transition the building permit process of the Banadir Regional Administration from a manual, paper-based workflow into an integrated,

automated software pipeline. System performance testing confirms that the application performs as expected and successfully addresses core administrative bottlenecks.

The integration of the multi-tier system optimizes operational workflows. Authorized field and verification inspectors utilize the responsive Flutter mobile app to access application records, verify document compliance, and approve structural statuses. Simultaneously, back-office municipal staff can review uploaded blueprints and cross-examine land ownership documentation in real time via the centralized web interface.

A key technical feature engineered into the core workflow is the dynamic financial calculation engine. When an application is returned for corrections by an inspector, the citizen is permitted to modify and resubmit their form without incurring an additional base application fee. However, if the applicant alters critical geometric parameters such as expanding the plot layout size, total building footprint area, or floor count the backend system automatically intercepts the data mutation, computes the dynamic fee difference, and prompts the user to settle the balance via the integrated payment gateway.

5.2.1 Comparison with Existing Local System and Municipal Context

To evaluate the practical contribution and operational impact of the **Urban Permit System**, its architecture was benchmarked against the existing permit issuance procedures currently utilized by the Banadir Regional Administration (BRA):

- **Elimination of In-Person Municipal Bureaucracy:** In the existing process, applicants are required to physically travel to local district offices to submit property details through an internal staff member. This model creates major delays due to office hour restrictions and physical unavailability. The developed system replaces this with a

24/7 mobile-first portal, allowing citizens to submit permit applications and blueprints remotely from anywhere.

- **Mitigation of Financial Leakage and Unofficial Receipting:** Under the current manual method, citizens transfer building fees directly to dynamic personal mobile numbers of local staff, who manually issue physical paper receipts. This practice exposes the municipality to financial opacity, delayed central account transfers, and potential revenue leakage. My platform integrates an automated payment gateway (WaafiPay) that processes payments directly to institutional municipal accounts and auto-generates tamper-proof digital receipts.
- **Replacement of Manual Field Verification:** In legacy operations, structural verification lacks a structured inspection protocol; field officers merely check physical paper slips on-site. The developed system introduces a digital inspector module, enabling authorized staff to perform structured digital verifications, log evaluation findings, and validate permit authenticity via system-generated QR codes.
- **End-to-End Workflow Transparency:** While existing district-level systems function in isolated silos without applicant visibility, the developed platform unifies citizens, district staff, inspectors, and super-administrators within a single Node.js and MongoDB pipeline, providing real-time application status tracking at every review stage.

5.3 Conclusion

The Building Permit System successfully achieved all primary research objectives established at the beginning of this project. The system provides a unified digital platform that modernizes municipal governance by replacing legacy manual routines for land

development auditing, paper-reliant permit requests, fee calculations, document management, and physical inspection tracking across Mogadishu Municipality.

The implementation of dynamic QR code verification significantly enhances municipal property regulation by allowing field inspectors to scan and verify construction permit authenticity on-site quickly and accurately. The digital application workspace simplifies the submission and auditing of property developments by enabling citizen applicants to submit structural dimensions, upload ownership deeds, calculate dynamic appraisal costs, and track application states (PENDING, APPROVED, RETURNED) electronically. Simultaneously, the web administration workspace improves municipal operational efficiency through centralized management of staff accounts, regional district consoles, building category parameters, application reviews, and real-time revenue analytics.

Furthermore, the system guarantees high operational reliability and data privacy through secure authentication, JWT stateless authorization, granular Role-Based Access Control (RBAC), bcrypt password encryption, and multi-tier input sanitization. Overall, the Building Permit System demonstrates that integrating modern web technologies (React.js), cross-platform mobile application frameworks (Flutter), RESTful API microservices (Node.js and Express.js), and flexible document database systems (MongoDB) can significantly transform public sector operations, eliminate bureaucratic delays, prevent revenue leakage, and establish transparent municipal governance in Mogadishu.

5.4 Limitations of the Study

Although the Building Permit System successfully achieved its core functional and architectural objectives, several technical and operational limitations were identified during the development and testing phases:

- **Network Connectivity Reliance:** The system architecture relies on active internet connectivity for real-time synchronization between the Flutter Mobile Client, React.js Web Admin Console, and Express.js REST APIs. Offline data caching and asynchronous queueing for field inspections in remote areas with limited network coverage are not fully supported.
- **Hardware & Environmental Scanning Constraints:** Camera-based QR code verification on construction sites depends on physical device camera quality, autofocus capabilities, and adequate ambient lighting conditions during field inspection cycles.
- **Manual Document & Spatial Verification:** Document auditing for property ownership deeds and identity cards relies on manual inspection by municipal staff officers via inline PDF/image modals, without integrating automated Optical Character Recognition (OCR) or AI-driven property boundary GIS spatial analysis.
- **Evaluation Environment Scope:** System testing was conducted within a simulated municipal environment for Mogadishu Municipality. Large-scale stress testing under concurrent high-volume production traffic across all national regional districts remains to be evaluated in future operational phases.

5.5 Recommendations for Future Work

Based on the empirical outcomes and operational limitations identified during the implementation and testing of the Building Permit System, several technical enhancements are recommended for future iterations:

- **Offline Inspection Caching:** Implementing local database caching (such as SQLite or Hive) on the mobile client will enable field inspectors to conduct camera-based QR code verifications and offline audit logging in areas with unstable internet connectivity, synchronizing records automatically once reconnected.
- **AI-Powered GIS & OCR Document Verification:** Integrating Artificial Intelligence, Optical Character Recognition (OCR), and Geographic Information System (GIS) mapping can automate land deed verification, detect document tampering, and validate property plot coordinates against official municipal spatial master plans.
- **Biometric & Multi-Factor Authentication:** Incorporating biometric authentication (fingerprint or facial recognition) for staff officers and field inspectors can strengthen system security and prevent credential-sharing during administrative approvals.
- **Municipal Payment Gateway & Land Registry Integration:** Expanding API integrations to connect directly with national banking systems, mobile money operators (EVC Plus, Sahal), and national land registry databases will allow automatic fee settlement verification and seamless cross-agency data validation.
- **Predictive Analytics & Executive Dashboards:** Developing advanced predictive analytics models can assist municipal urban planners in forecasting construction density, identifying high-growth district zones, and optimizing municipal infrastructure resource allocation.

5.6 Final Remarks

The Building Permit System provides a practical, secure, and reliable digital solution designed to modernize building permit governance and streamline property auditing across Mogadishu Municipality. By integrating a responsive React.js web admin console, a cross-platform Flutter mobile application, camera-based QR code verification, and a secure Node.js/MongoDB backend, the system demonstrates the powerful potential of digital transformation in municipal administration.

The successful implementation and verification of this project establish a strong foundation for future technological adoption and policy-driven digitalization in public service delivery. With continuous maintenance, institutional support, and future feature expansions, the platform can evolve into a comprehensive urban governance system capable of supporting the growing infrastructure needs of Mogadishu Municipality.

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